

Domestic Intermediate Input Access and Multiproduct Firm Exports*

Longfei Gao[†], Yao Tang[‡]

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Abstract

We study how access to domestic intermediate inputs affects firms' export performance. To this end, we develop a multiproduct-firm model with domestic intermediate-input trade. Improved access to intermediate goods generates both cost-saving and competition effects. The cost-saving effect dominates at the extensive margin, leading firms to expand their export product scope, while the effects on product-level exports are heterogeneous: lower-markup products benefit more from improved access to intermediate inputs, whereas higher-markup products are more adversely affected by intensified competition. We test these predictions using firm-level data on Chinese manufacturing firms during 2000-2007 and a novel upstream market access (UMA) measure that captures firms' access to domestic intermediate suppliers. Using a combined instrumental-variables and control-function strategy, we find that a one-percent increase in city-industry-level UMA raises firm exports by 0.083%. The increase in exports is accompanied by a robust expansion of export scope, as the number of export varieties increases by 0.032%, while positive effects on the intensive margin are present but less uniform across measures. Consistent with the model, improvements in UMA have more favorable effects on product-level exports for firms with lower markups. We further show that higher UMA reduces firms' intermediate-input costs while increasing net firm entry, providing evidence for the cost-saving and competition mechanisms highlighted by the model.

JEL classification: F10, F15, R4

Keywords: domestic intermediate inputs; market access; multiproduct firms; export margins; railway infrastructure

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[†]Department of Applied Economics, Guanghua School of Management, Peking University, 5 Yihuayuan Road, Haidian District, Beijing 100871, China. E-mail: jimfmgao@pku.edu.cn.

[‡]Corresponding author. Department of Applied Economics, Guanghua School of Management, and Institute of Carbon Neutrality, Peking University, 5 Yihuayuan Road, Haidian District, Beijing 100871, China. E-mail: ytang@gsm.pku.edu.cn.

1 Introduction

China’s rapid export expansion following its accession to the World Trade Organization (WTO) in 2001 coincided with a period of deepening domestic market integration. While a large literature has examined how exports respond to trade liberalization, foreign market access, and transportation infrastructure, much less attention has been paid to the role of access to domestic intermediate inputs. This omission is notable because intermediate goods account for a substantial share of firms’ production costs, and improvements in domestic transportation networks can fundamentally reshape firms’ access to suppliers. By lowering the cost of sourcing intermediate inputs from geographically distant locations, domestic market integration may affect firms’ export performance even when transportation costs for final goods remain unchanged.

In this paper, we examine how improved access to domestic intermediate inputs affects the export performance of Chinese manufacturing firms. To guide the empirical analysis, we extend the multiproduct-firm framework of Mayer, Melitz, and Ottaviano (2014) by incorporating domestic intermediate-input trade and a spatial structure consisting of a seaboard and an inland location in each country. The model highlights two opposing forces generated by reductions in domestic trade costs. Improved access to intermediate goods lowers production costs, while increased market integration intensifies competition among firms. We show that the cost-saving effect dominates at the extensive margin, leading firms to expand their export product scope. Along the intensive margin, however, the effects on product-level exports are heterogeneous. Lower-markup varieties benefit more from improved access to intermediate inputs, while higher-markup varieties are more exposed to intensified competition. As a result, firm-level exports increase when the expansion of export varieties and the growth of exports among lower-markup products outweigh the negative effects on higher-markup products.

To measure firms’ access to domestic intermediate goods, we combine firm-level

data on Chinese manufacturing firms during 2000–2007 with information on the evolution of China’s railway network. We construct an Upstream Market Access (UMA) index that incorporates inter-industry input-output linkages into the traditional market-access framework. The index captures improvements in firms’ access to domestic suppliers of intermediate goods arising from railway expansion.

Identifying the causal effect of UMA is challenging because railway expansion is unlikely to be randomly assigned across locations. To address this issue, we adopt a combined instrumental-variables and control-function strategy following Borusyak and Hull (2023). Drawing on Faber (2014) and Banerjee, Duflo, and Qian (2020), we construct two artificial transportation networks based on geographic and historical factors and use the resulting counterfactual UMA measures as instruments for realized UMA. In addition, we control for firms’ expected upstream market access (EUMA) in the second-stage regression to account for the possibility that geographically advantaged locations are systematically more likely to be connected by railways. This strategy allows us to identify the effect of exogenous changes in access to domestic intermediate inputs on firms’ export behavior.

Empirically, we find that a one-percent increase in city-industry-level UMA raises firm-level exports by 0.083%. The effect operates primarily through the extensive margin: the number of export varieties increases by 0.032%, and alternative measures of export scope exhibit similar responses. We also find evidence of positive effects along certain dimensions of the intensive margin. Consistent with the model’s predictions, improvements in upstream market access have more favorable effects on product-level exports for firms with lower markups. Finally, we provide evidence on the underlying mechanisms. Higher UMA significantly reduces firms’ intermediate-input costs while simultaneously increasing net firm entry, indicating that improved access to domestic intermediate goods generates both cost-saving and competition effects.

This paper contributes to the literature on transportation infrastructure and trade

by identifying access to domestic intermediate inputs as an important mechanism linking domestic market integration to export performance. Previous studies have shown that transportation infrastructure promotes trade at both the regional level (Duranton, Morrow, and Turner, 2014; Coşar and Demir, 2016; Donaldson, 2018) and the firm level (Martincus and Blyde, 2013; Martincus, Carballo, and Cusolito, 2017; Fan, Lu, and Luo, 2023; Liu, Sheng, and Yu, 2023). Most of this literature emphasizes the role of transportation costs for final goods. In contrast, we focus on how domestic transportation infrastructure affects firms’ access to intermediate suppliers. Our paper is particularly related to Donaldson (2018), who shows that railroad expansion facilitated both inter-regional and international trade in colonial India. We complement this literature by showing that improved access to domestic intermediate inputs increases firm exports through both extensive- and intensive-margin adjustments and by documenting the underlying cost-saving and competition mechanisms.

This paper also contributes to the literature on export margins and multiproduct firms. At the aggregate level, Amiti and Freund (2008) show that the intensive margin accounted for most of China’s export expansion following WTO accession. At the firm level, a large literature has examined how factors such as credit access (Manova, Wei, and Zhang, 2015), road infrastructure (Fan, Lu, and Luo, 2023; Liu, Sheng, and Yu, 2023), and high-speed railways (Tang et al., 2019; Kee, Xie, and Xu, 2024) affect firms’ export margins. We contribute to this literature in two ways. First, we provide evidence that improved access to domestic intermediate inputs is an important determinant of firms’ export scope. Second, by embedding intermediate-input trade into the multiproduct-firm framework of Mayer, Melitz, and Ottaviano (2014), we show that reductions in domestic trade costs generate heterogeneous effects on product-level exports, with more favorable responses among lower-markup firms. These heterogeneous product-level responses help explain how domestic market integration translates into firm-level export growth.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and derives the main predictions regarding domestic intermediate-input access and firm exports. Section 3 describes the data, variable construction, and identification strategy. Section 4 presents the empirical evidence and investigates the cost-saving and competition mechanisms. Section 5 concludes.

2 Model

In a multiproduct-firm model following Mayer, Melitz, and Ottaviano (2014), we introduce domestic intermediate-input trade and examine how improved access to intermediate goods affects firms' export performance. Changes in domestic trade costs matter because they alter firms' access to upstream suppliers and the intensity of competition they face.

2.1 Closed Economy

Preference. Consider an economy with L consumers. They consume a homogeneous good that serves as the numéraire and a continuum of differentiated goods. All consumers have identical preferences described by the following utility function:

$$U = q_0^c + \alpha \int_{i \in \Omega} q_i^c di - \frac{1}{2} \gamma \int_{i \in \Omega} (q_i^c)^2 di - \frac{1}{2} \eta \left(\int_{i \in \Omega} q_i^c di \right)^2, \quad (1)$$

where q_0^c and q_i^c represent the individual consumption of the numéraire good and variety i of differentiated goods from the set Ω . The demand parameters α , γ , and η are all positive. The parameters α and η measure the degree to which consumers prefer the differentiated varieties relative to the numéraire. The parameter γ the degree of substitutability across varieties. A larger γ implies a lower degree of substitutability across varieties, and consumers place greater weight on smoothing consumption across varieties.

Each consumer supplies one unit of labor and makes consumption decisions subject to a budget constraint. The inverse demand for variety i is given by:

$$p_i = \alpha - \gamma q_i^c - \eta Q^c, \quad (2)$$

where p_i is the price of variety i and $Q^c = \int_{i \in \Omega} q_i^c di$ is the total demand for the differentiated products. It can be shown that the linear market demand function for variety i is

$$q_i \equiv Lq_i^c = \frac{\alpha L}{\eta\mu + \gamma} - \frac{L}{\gamma} p_i + \frac{\eta\mu L}{(\eta\mu + \gamma)\gamma} \bar{p}, \quad \forall i \in \Omega^*, \quad (3)$$

where μ is the measure of varieties in the subset $\Omega^* \subset \Omega$ that are consumed (with positive q_i^c).¹ $\bar{p} = \frac{1}{\mu} \int_{i \in \Omega^*} p_i di$ is the average price over Ω^* . The demand elasticity here depends on the intensity of market competition, which is associated with the degree of substitutability across varieties (γ), the measure of competing varieties (μ), and their average price ($\bar{p}$).²

Production. Let the differentiated final products of a firm be indexed by the non-negative integer m , where the productivity of variety m is denoted by φ_m . The production of each variety requires two inputs: labor (L) and a composite of intermediate goods (X), both of which are assumed to be inelastically supplied in a competitive market. The production technology for each variety is Cobb-Douglas:

$$q = \varphi_m L^\beta X^{1-\beta}, \quad (4)$$

From the firm's cost minimization problem, the unit production cost of a variety is:³

$$c = \frac{1}{\varphi_m} w^\beta p_X^{1-\beta}, \quad (5)$$

where w and p_X denote the prices of labor and the composite intermediate good, respectively.

Following Mayer, Melitz, and Ottaviano (2014), each firm possesses a core variety that represents its “core competency.” The core variety is indexed by $m = 0$, while

¹The upper price bound $p^{max} \equiv \frac{\alpha\gamma + \eta\mu\bar{p}}{\eta\mu + \gamma}$ is defined by letting $q_i = 0$ in Equation (3). Therefore, all varieties $i \in \Omega^*$ with positive demand must satisfy $p_i < p^{max}$.

²In contrast, the demand elasticity under the CES preference is constant given the degree of product differentiation.

³The exact expression from cost minimization is $c = \frac{1}{\beta^\beta (1-\beta)^{1-\beta}} \frac{1}{\varphi_{min}} w^\beta p_X^{1-\beta}$, which equals Equation (5) multiplied by the constant $\frac{1}{\beta^\beta (1-\beta)^{1-\beta}}$. With an appropriate normalization of φ_{min} , one obtains the simplified expression in Equation (5).

varieties with larger values of m are progressively farther from the firm's core competency. For a firm whose core variety has productivity φ , the productivity of variety m is

$$\varphi_m = \omega^m \varphi_0, \quad 0 < \omega < 1.$$

The parameter ω governs the efficiency gap between varieties. As ω approaches one, firms are more likely to expand their product range because varieties farther from the core competency incur relatively small efficiency losses. Conversely, as ω approaches zero, firms are more likely to produce only their core variety.

To enter the market, a final-good firm must incur a fixed entry cost f_E before observing the productivity of its core variety φ_0 . We assume that φ_0 is drawn from a common distribution $G_\varphi(\varphi)$ with support $[\varphi_{min}, \infty)$.

Equivalently, we can characterize the core competency in terms of production cost. Conditional on factor prices as in (5), the unit cost c follows the distribution $G(c)$ with support $(0, \bar{c}]$, where $\bar{c} = \frac{1}{\varphi_{min}} w^\beta p_X^{1-\beta}$. Define $c_0 = \frac{1}{\varphi_0} w^\beta p_X^{1-\beta}$, i.e., c_0 denotes the unit cost of the core product of a firm with core productivity φ_0 . With the notation, the unit cost of the m -th variety of a firm with core productivity φ_0 can be written as

$$v(m, c_0) = \frac{1}{\varphi_m} w^\beta p_X^{1-\beta} = \omega^{-m} c_0 \quad (6)$$

As m increases, i.e., as a firm's production moves away from its core competency, the unit cost $v(m, c_0)$ increases.

Firm Behavior. After incurring the (sunk) entry cost, a firm then decides on the production of each variety. Product m will be produced if the firm can cover its unit cost $v(m, c_0)$. Note that a firm survives only if it earns a non-negative profit on its core variety. Otherwise, the firm will immediately exit the market. Specifically, given a core unit cost c_0 , a surviving firm will maximize the following operating profit from the core variety:

$$\max_{p(c_0), q(c_0)} \pi(c_0) = (p(c_0) - c_0)q(c_0), \quad (7)$$

where $q(c_0) = \frac{\alpha L}{(\eta\mu + \gamma)} - \frac{L}{\gamma}p(c_0) + \frac{\eta\mu L}{(\eta\mu + \gamma)\gamma}\bar{p}$. Define

$$c_M = (\alpha\gamma + \eta\mu\bar{p})/(\eta\mu + \gamma)$$

as the survival cutoff cost, namely the highest unit cost at which a firm can earn non-negative profits. Only those firms with $c_0 \leq c_M$ could survive and produce their core variety. The production choice for other varieties is similar: a firm could profitably produce their m -th variety if $v(m, c_0) \leq c_M$, that is, $c_0 \leq \omega^m c_M$. Therefore, the production scope of a firm with unit cost c_0 , defined by the number of varieties it produces, is given by:

$$M(c_0) = \begin{cases} \max(m \mid c_0 \leq \omega^m c_M) + 1 & \text{if } c_0 \leq c_M \\ 0 & \text{if } c_0 > c_M. \end{cases} \quad (8)$$

It follows that $M'(c_0) \leq 0$ and $\partial M(c_0)/\partial c_M \geq 0$, i.e., firms with lower core unit costs (due to higher productivity φ_0) could supply more varieties in markets, and markets with less intensive competition (represented by a higher c_M) can accommodate more varieties from a firm. For each variety m produced by a firm with core unit cost c_0 , the equilibrium outcomes can be expressed in terms of c_M and c_0 :

$$q(m, c_0) = \frac{L}{2\gamma}(c_M - v(m, c_0)) = \frac{L}{2\gamma}(c_M - \omega^{-m}c_0), \quad (9)$$

$$p(m, c_0) = \frac{1}{2}(c_M + v(m, c_0)) = \frac{1}{2}(c_M + \omega^{-m}c_0), \quad (10)$$

$$\lambda(m, c_0) = \frac{1}{2}(c_M - v(m, c_0)) = \frac{1}{2}(c_M - \omega^{-m}c_0), \quad (11)$$

$$r(m, c_0) = \frac{L}{4\gamma}(c_M^2 - v(m, c_0)^2) = \frac{L}{4\gamma}(c_M^2 - (\omega^{-m}c_0)^2), \quad (12)$$

$$\pi(m, c_0) = \frac{L}{4\gamma}(c_M - v(m, c_0))^2 = \frac{L}{4\gamma}(c_M - \omega^{-m}c_0)^2, \quad (13)$$

where $\lambda(m, c_0) = p(m, c_0) - v(m, c_0)$ and $r(m, c_0) = p(m, c_0)q(m, c_0)$ are the markup and revenue of variety m . From the above results, firms with a lower c_0 will set lower prices and face higher market demand, thus yielding higher profits. Meanwhile, as an indicator of tougher market competition, a decrease in c_M reduces the prices, output, and profits for all varieties that are produced.

After entering the market, the total profit of a firm with a core unit cost c_0 can be derived from Equations (8) and (13):

$$\Pi(c_0) = \sum_{m=0}^{M(c_0)-1} \pi(m, c_0) = \sum_{m=0}^{M(c_0)-1} \frac{L}{4\gamma} (c_M - \omega^{-m} c_0)^2. \quad (14)$$

Before entering the market, the expected profit for a firm is $\int_0^{c_M} \Pi(c_0) dG(c_0) - f_E$, and hence the free entry equilibrium condition is:

$$\int_0^{c_M} \Pi(c_0) dG(c_0) = f_E. \quad (15)$$

To keep the subsequent analysis tractable, we parameterize the distribution of the firm's core productivity φ_0 . Specifically, φ is assumed to follow a Pareto distribution with a lower bound φ_{min} and a shape parameter $\xi \geq 1$, i.e., $G_\varphi(\varphi_0) = 1 - (\frac{\varphi_{min}}{\varphi_0})^\xi$. The implied distribution of unit cost c_0 is:

$$G(c_0) = (\frac{c_0}{\bar{c}})^\xi, \quad c_0 \in [0, \bar{c}]. \quad (16)$$

The parameter ξ measures the degree of cost dispersion. With a larger ξ , the cost distribution is more concentrated at higher costs.

Under the assumption of a Pareto distribution of costs, we can derive from the free entry condition the analytical solution for the cutoff cost c_M and the mass of surviving firms N in equilibrium (see the Appendix for the proof):

$$\begin{aligned} c_M &= \left(\frac{\gamma\theta}{L\zeta} \right)^{\frac{1}{\xi+2}}, \\ N &= \frac{2(\xi+1)\gamma}{\eta\zeta} \frac{\alpha - c_M}{c_M}, \end{aligned} \quad (17)$$

where $\theta = 2(\xi+1)(\xi+2)(\bar{c})^\xi f_E$ and $\zeta = \frac{1}{1-\omega^\xi} > 1$. The equilibrium c_M is negatively correlated with the local market size L , while positively correlated with entry costs f_E and the upper bound of unit cost $\bar{c}$.

2.2 Open Economy

In this section, we will derive the equilibrium in an open economy model with trade costs. In the model, domestic trade costs influence firms primarily through their access to intermediate inputs. For completeness, they also affect the transportation costs of final goods. We will discuss the implications of the reduction in trade costs in a simplified trade structure in Section 2.3.

We first consider a general open economy consisting of multiple countries, indexed by $j = 1, 2, \dots, J$. Each country contains cities that are indexed by $f = 1, 2, \dots, F_j$. Let jf denote city f in country j . There is an iceberg trade cost $\tau_{jfk g}$ between city jf and kg , i.e., for $e^{\tau_{jfk g}} > 1$ units of final goods shipped from kg , only one unit arrives at jf .⁴ Assume that trade costs are symmetric such that $\tau_{jfk g} = \tau_{kgj f}$. When $jf = kg$, it holds that $e^{\tau_{jfk g}} = 1$.

The price of intermediate goods is country-specific and it is affected by trade costs. Specifically, the CES price index of intermediate goods for country j is:

$$p_{Xj} = \left(\sum_{k=1}^J (e^{\tau_{jk}} p_{I,k})^\sigma \right)^{\frac{1}{\sigma}}, \quad (18)$$

where $p_{I,k}$ is the input cost in country k , and $e^{\tau_{jk}}$ represents the international trade cost between countries j and k .⁵ Furthermore, the input cost is a function of domestic trade costs, i.e., $p_{I,k} = p_{I,k}(T_k)$, where $T_k = (\{e^{\tau_{ik}}\})$ is a matrix of trade costs between each pair of cities within country k .

Similar to the setup in a closed economy, the productivity of a firm's core product follows the distribution $G_\varphi(\varphi)$ with support $[\varphi_{min}, \infty]$. Note that the unit cost of core competence now has a country-specific upper bound $\bar{c}_j = \frac{1}{\varphi_{min}} w^\beta p_{Xj}^{1-\beta}$ for firms in country j .

⁴For small values of τ , it holds that $e^\tau \approx 1 + \tau$.

⁵It is derived from a production function for intermediate goods with CES technology $X_j = (\sum_{k=1}^J I_k^\epsilon)^{\frac{1}{\epsilon}}$, where I_i is the raw input from country k and $\epsilon > 0$ is a parameter of substitution. $e^{\tau_{jk}}$ can be considered as the trade cost between the ports of two countries, which will be discussed in detail in Section 2.3.

For any city jf , we first determine the cutoff cost for firms to survive in the local market, denoted by $c_{M,jfjf} = \frac{\alpha\gamma + \eta\mu_{jf}\bar{p}_{jf}}{\eta\mu_{jf} + \gamma}$, where μ_{jf} is the measure of varieties sold in city jf , and $\bar{p}_{jf}$ is the average price. Due to the existence of trade costs, for a firm in city kg with a core unit cost c_0 , the total cost of selling in city jf is $c_0 e^{\tau_{jfk g}}$. Therefore, the cutoff cost for exporting from kg to jf is $c_{M,jfkg} = \frac{c_{M,jfjf}}{e^{\tau_{jfk g}}}$.

The number of varieties a firm with cost c_0 exports from kg to jf is given by:

$$M_{jfk g}(c_0) = \begin{cases} \max(m \mid c_0 \leq \omega^m c_{M,jfkg}) + 1 & \text{if } c_0 \leq c_{M,jfkg} \\ 0 & \text{if } c_0 > c_{M,jfkg} \end{cases} \quad (19)$$

When $kg = jf$, this equation represents the production scope for the local market. Again, we have $M'_{jfk g}(c_0) \leq 0$ and $\partial M_{jfk g}(c_0) / \partial c_{M,jfkg} \geq 0$. Namely, firms with lower core unit costs could supply more varieties to markets; firms can sell more varieties to markets with less intense competition.

After a firm from country-city kg enters market jf , the product-level equilibrium outcomes are:

$$q_{jfk g}(m, c_0) = \frac{L_{jf}}{2\gamma} e^{\tau_{jfk g}} (c_{M,jfkg} - v(m, c_0)) = \frac{L_{jf}}{2\gamma} (c_{M,jfjf} - e^{\tau_{jfk g}} \omega^{-m} c_0), \quad (20)$$

$$p_{jfk g}(m, c_0) = \frac{1}{2} e^{\tau_{jfk g}} (c_{M,jfkg} + v(m, c_0)) = \frac{1}{2} (c_{M,jfjf} + e^{\tau_{jfk g}} \omega^{-m} c_0), \quad (21)$$

$$\lambda_{jfk g}(m, c_0) = \frac{1}{2} e^{\tau_{jfk g}} (c_{M,jfkg} - v(m, c_0)) = \frac{1}{2} (c_{M,jfjf} - e^{\tau_{jfk g}} \omega^{-m} c_0), \quad (22)$$

$$r_{jfk g}(m, c_0) = \frac{L_{jf}}{4\gamma} (e^{\tau_{jfk g}})^2 (c_{M,jfkg}^2 - v(m, c_0)^2) = \frac{L_{jf}}{4\gamma} (c_{M,jfjf}^2 - (e^{\tau_{jfk g}} \omega^{-m} c_0)^2), \quad (23)$$

$$\pi_{jfk g}(m, c_0) = \frac{L_{jf}}{4\gamma} (e^{\tau_{jfk g}})^2 (c_{M,jfkg} - v(m, c_0))^2 = \frac{L_{jf}}{4\gamma} (c_{M,jfjf} - e^{\tau_{jfk g}} \omega^{-m} c_0)^2, \quad (24)$$

For the firm, the profits in the local and export markets are:

$$\Pi_{kgkg}(c_0) = \sum_{m=0}^{M_{kgkg}(c)-1} \pi_{jfk g}(m, c_0) = \sum_{m=0}^{M_{kgkg}(c)-1} \frac{L^{kg}}{4\gamma} (c_{M,kgkg} - \omega^{-m} c_0)^2, \quad (25)$$

$$\Pi_{jfk g}(c_0) = \sum_{m=0}^{M_{jfk g}(c)-1} \pi_{jfk g}(m, c_0) = \sum_{m=0}^{M_{jfk g}(c)-1} \frac{L_{jf}}{4\gamma} (c_{M,jfjf} - e^{\tau_{jfk g}} \omega^{-m} c_0)^2. \quad (26)$$

The free entry condition for a potential entrant at city kg is:

$$\sum_{j=1}^J \sum_{f=1}^{F_j} \int_0^{c_{M,jfkg}} \Pi_{jfg}(c_0) dG_{c,k}(c_0) = f_E, \quad (27)$$

where the left side is the sum of expected profit in the local market kg and all exporting markets, in which $G_{c,k}(c_0)$ is the distribution of core-cost draws for firms located in country k . The right side is the fixed entry cost.

For country $j = 1, 2, \dots, J$ and city $f = 1, 2, \dots, F_j$, there are $\sum_{j=1}^J F_j$ free entry conditions that form a system of equations. Applying Cramer's rule, the cutoff cost of serving the local market for each city can be solved as:⁶

$$c_{M,jfjf} = \left(\frac{\gamma}{L_{jf}\zeta} \frac{|\mathcal{P}_{jf}^{\theta}|}{|\mathcal{P}|} \right)^{\frac{1}{\xi+2}}. \quad (28)$$

where L_{jf} is the market size of city jf , and $\theta_j = 2(\xi + 1)(\xi + 2)f_E(\bar{c}_j)^\xi$. $|\mathcal{P}|$ is the determinant of the following matrix:

$$\mathcal{P} = \begin{pmatrix} \rho_{11} & \rho_{12} & \dots & \rho_{1J} \\ \rho_{21} & \rho_{22} & \dots & \rho_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{J1} & \rho_{J2} & \dots & \rho_{JJ} \end{pmatrix}, \quad (29)$$

in which ρ_{jk} is a matrix of inverse trade costs between cities in countries j and k . Specifically, ρ_{jk} takes the form:

$$\rho_{jk} = \begin{pmatrix} \rho_{j1k1} & \rho_{j1k2} & \dots & \rho_{j1kF_k} \\ \rho_{j2k1} & \rho_{j2k2} & \dots & \rho_{j2kF_k} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{jF_jk1} & \rho_{jF_jk2} & \dots & \rho_{jF_jkF_k} \end{pmatrix}, \quad (30)$$

where $\rho_{jfg} \equiv \frac{1}{(e^{\tau_{jfg}})^{\xi}} \leq 1$ is negatively correlated with the trade cost $e^{\tau_{jfg}}$ and hence measures the market integration between kg and jf .⁷ Note that $\rho_{jfg} = 1$ for $jf = kg$.

⁶See Appendix for details on the solution.

⁷It is referred to as "freeness" of trade in Mayer, Melitz, and Ottaviano (2014).

$|\mathcal{P}_{jf}^\theta|$ in Equation (28) is the determinant of the matrix obtained by replacing the column of $\mathcal{P}$ corresponding to city jf with the vector θ :⁸

$$\theta \equiv \left(\underbrace{\theta_1, \dots, \theta_1}_{F_1}, \underbrace{\theta_2, \dots, \theta_2}_{F_2}, \dots, \underbrace{\theta_J, \dots, \theta_J}_{F_J} \right)^\top. \quad (31)$$

The cutoff cost of survival for exporters from kg to jf is:

$$c_{M,jfkg} = \left[\frac{\gamma}{L_{jf}\zeta} \frac{|\mathcal{P}_{jf}^\theta|}{|\mathcal{P}|} (\rho_{jfk g})^{\frac{\xi+2}{\xi}} \right]^{\frac{1}{\xi+2}}. \quad (32)$$

The open-economy model presented above provides the general framework of the analysis. To derive tractable comparative statics and isolate the role of domestic intermediate-input access, we next consider a more structured environment in which the effects of changes in domestic trade costs can be characterized analytically.

2.3 An Inland-Seaboard Trade Structure

We specialize the model to a 2×2 geographic structure consisting of two countries, $j = 1, 2$, each with two cities. This parsimonious setting allows us to characterize analytically how changes in domestic trade costs—and hence access to domestic intermediate goods—affect firms' export behavior.

We furthermore assume that in each country, one city is land-locked (indexed by l) and the other is on the seaboard (indexed by s). As in Coşar and Fajgelbaum (2016), we assume that only the seaboard city can trade directly with the other country. That is, all traded goods must pass through a port to be exported to international markets or imported into domestic markets. As described in Figure 1, the domestic trade cost between cities jl and js in country j ($j = 1, 2$) is denoted by τ_j , and the international trade cost between 1 and 2 is τ_0 . It means, for example, exporting products from $1l$ to $2l$ incurs an iceberg trade cost $e^{\tau_1 + \tau_0 + \tau_2}$, which includes two segments of domestic trade costs (τ_1 and τ_2) and one part of international trade costs (τ_0).

⁸Note that all cities within country j share the same cost-distribution parameter θ_j .

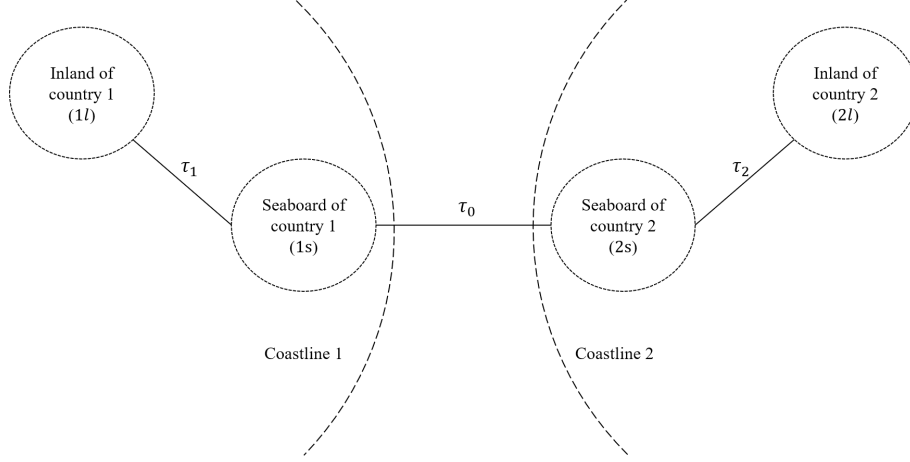


Figure 1: An Inland-Seaboard Trade Structure

From Equation (18), the prices of intermediate goods in countries 1 and 2 are given by:

$$P_{X,1} = (p_{I,1}^\sigma + (e^{\tau_0} p_{I,2})^\sigma)^{\frac{1}{\sigma}}, \quad (33)$$

$$P_{X,2} = (p_{I,2}^\sigma + (e^{\tau_0} p_{I,1})^\sigma)^{\frac{1}{\sigma}}, \quad (34)$$

where $p_{I,j} = p_{I,j}(e^{\tau_j})$, $j = 1, 2$.

Under this geography, we can rewrite the cutoffs to survive in the local markets of jl and js , $j = 1, 2$:

$$c_{M,1s1s} = \left(\frac{\gamma}{L_{1s}\zeta} \frac{\kappa_1}{(1 - \rho_0^2)(1 + \rho_1)} \right)^{\frac{1}{\xi+2}}, \quad (35)$$

$$c_{M,1l1l} = \left(\frac{\gamma}{L_{1l}\zeta} \frac{\theta_1}{1 + \rho_1} \right)^{\frac{1}{\xi+2}}, \quad (36)$$

$$c_{M,2s2s} = \left(\frac{\gamma}{L_{2s}\zeta} \frac{\kappa_2}{(1 - \rho_0^2)(1 + \rho_2)} \right)^{\frac{1}{\xi+2}}, \quad (37)$$

$$c_{M,2l2l} = \left(\frac{\gamma}{L_{2l}\zeta} \frac{\theta_2}{1 + \rho_2} \right)^{\frac{1}{\xi+2}}. \quad (38)$$

where $\kappa_1 = \theta_1(1 + \rho_0^2\rho_1) - \theta_2\rho_0(1 + \rho_1)$, $\kappa_2 = \theta_2(1 + \rho_0^2\rho_2) - \theta_1\rho_0(1 + \rho_2)$, $\rho_i = (e^{\tau_i})^{-\xi} < 1$ for $i = 0, 1, 2$, and $\theta_j = 2(\xi + 1)(\xi + 2)f_E(w^\beta P_{X,j}^{1-\beta}/\varphi_{min})^\xi$ for $j = 1, 2$. From $c_{M,1s1s} > 0$ and $c_{M,2s2s} > 0$, we have $\frac{1}{e^{\tau_0}} < \frac{P_{X,1}}{P_{X,2}} < e^{\tau_0}$.

We now analyze how changes in domestic trade costs affect firms through changes in access to domestic intermediate inputs. Due to the symmetry in the equilibrium solution between the two countries, we take country 1 as the home country in the following analysis and discuss the impact of domestic trade cost τ_1 .

Define

$$\begin{aligned} a_1 &\equiv (1 - \beta) \frac{\partial \ln P_{X,1}}{\partial \tau_1} = (1 - \beta) \left(\frac{p_{I,1}}{P_{X,1}} \right)^\sigma \frac{\partial \ln p_{I,1}}{\partial \tau_1} > 0, \\ \chi &\equiv \frac{\partial \ln P_{X,2}/\partial \tau_1}{\partial \ln P_{X,1}/\partial \tau_1} = \left(\frac{e^{\tau_0} P_{X,1}}{P_{X,2}} \right)^\sigma \end{aligned} \quad (39)$$

where $0 < \chi < 1$ under the maintained CES parameterization.

In this model, a reduction in domestic trade costs τ_1 affects firms primarily through changes in access to domestic intermediate goods. Improved access generates two distinct effects. First, lower prices of intermediate goods reduce firms' core production costs for a given level of productivity, which is reflected by⁹

$$\partial c_{0,1f}/\partial \tau_1 = a_1 c_{0,1f} > 0, \quad f = s, l.$$

We refer to this as the *cost-saving effect of intermediate goods*.

Second, lower intermediate-input prices alter the cost distribution of entrant firms and therefore affect market competition. Formally,¹⁰

$$\partial \bar{c}_1/\partial \tau_1 = a_1 \bar{c}_1 > 0$$

$$\partial \bar{c}_2/\partial \tau_1 = \chi a_1 \bar{c}_2 > 0.$$

We refer to this as the *competition effect of intermediate goods*. These two effects constitute the central mechanism of the paper.

For completeness, a reduction in τ_1 also directly lowers the transportation cost of final goods within country 1. We refer to this channel as the *final-goods transportation*

⁹It follows that $\partial p_{I,1}/\partial \tau_1 = e^{\tau_1} p_{X,1}^{1-\sigma} p_{I,1}^\sigma \frac{\partial p_{I,1}/\partial \tau_1}{p_{I,1}}$.

¹⁰Since θ_j is proportional to $\bar{c}_j^\xi$, we also have $\frac{\partial \ln \theta_1}{\partial \tau_1} = \xi a_1$ and $\frac{\partial \ln \theta_2}{\partial \tau_1} = \xi \chi a_1$.

effect. Lower transportation costs improve a firm's price competitiveness in external markets while simultaneously exposing local firms to stronger competition from final goods produced in other cities. The effects of transportation infrastructure operating through final-goods transportation have been widely studied in the existing literature. In contrast, the effects operating through access to domestic intermediate goods have received considerably less attention. In the empirical analysis, we separately control for transportation costs of final goods to isolate the effects operating through access to domestic intermediate goods.

2.3.1 Domestic Trade Cost and Cutoff Cost for Firm Survival

Differentiate the cutoff cost for firms to survive in the local markets $1s$ and $1l$ with respect to τ_1 , we have:

$$\frac{\partial c_{M,1s1s}}{\partial \tau_1} = \frac{\xi}{\xi + 2} \frac{c_{M,1s1s}}{\kappa_1} (a_1 \kappa_3 + \kappa_4), \quad (40)$$

$$\frac{\partial c_{M,1l1l}}{\partial \tau_1} = \frac{\xi}{\xi + 2} c_{M,1l1l} \left(a_1 + \frac{\rho_1}{1 + \rho_1} \right) > 0, \quad (41)$$

$$(42)$$

where $\kappa_3 \equiv \theta_1(1 + \rho_0^2 \rho_1) - \chi \theta_2 \rho_0(1 + \rho_1)$ and $\kappa_4 = \frac{\theta_1 \rho_1(1 - \rho_0^2)}{1 + \rho_1} > 0$.

In each of the two equations above, the first term in the square bracket corresponds to the competition effect of intermediate goods, and the second term represents the final goods competition effect. The final goods competition effect always leads to tougher local competition for both city $1l$ and $1s$, because final goods shipped from the other domestic city become more competitive due to reduced transportation costs.

The competition effect of intermediate goods is more complex. Lower domestic input costs decrease the prices of both domestic and foreign intermediate goods. When the effect of the decline in intermediate goods prices domestically is greater, the cost distribution of entrant firms in country 1 will be shifted to the left, which intensifies competition in cities of country 1. Otherwise, the cost savings of intermediate goods will increase

the cutoff of the domestic market. In other words, the impact of the competition effect of intermediate goods on the cutoff cost depends on the redistribution of entrant firms' shares from different countries at a given market. For city 1s, this effect is ambiguous because, as a seaboard city, it is more exposed to foreign competition.¹¹ City 1l, however, only experiences increased local competition due to changes in domestic intermediate goods prices. This is because, in inland areas, domestic firms hold a greater advantage, and the market pattern is primarily shaped by domestic entrants. As a result, when trade costs τ_1 decrease, the local cutoff of 1l decreases, while that of 1s depends on the combination of the final-goods transportation effect and competition effect of intermediate goods.

Similarly, the derivatives of $c_{M,2s2s}$ and $c_{M,2l2l}$ with respect to τ_1 are:

$$\frac{\partial c_{M,2s2s}}{\partial \tau_1} = \frac{\xi}{\xi + 2} a_1 \frac{\kappa_5}{\kappa_2} c_{M,2s2s} > 0, \quad (43)$$

$$\frac{\partial c_{M,2l2l}}{\partial \tau_1} = \frac{\xi}{\xi + 2} \chi a_1 c_{M,2l2l} > 0, \quad (44)$$

where $\kappa_5 = \chi \theta_2 (1 + \rho_0^2 \rho_2) - \theta_1 \rho_0 (1 + \rho_2) > 0$.¹² Note that the decrease in trade costs in country 1 only affects the local cutoffs in country 2 through the competition effect of intermediate goods.¹³ Similar to the results of country 1, this effect depends on the relative changes in intermediate goods prices between the two countries. Differently, the change in country 2's intermediate goods price index dominates and hence always brings stronger competition to both 2s and 2l.

Based on the discussion above, we have the following lemma about local competition:¹⁴

Lemma 1. *A reduction in domestic trade costs in country 1 will: (i) increase cutoff costs in domestic inland city 1l and foreign cities 2s and 2l through the competition effect of*

¹¹Specifically, $\kappa_3 > 0$ if $\frac{p_{X1}}{p_{X2}} < \left(\frac{1 + \rho_0^2 \rho_1}{e^{\tau_0 \sigma} \rho_0 (1 - \rho_1)} \right)^{\frac{1}{\sigma + \beta - 1}}$, in which case the competition effect of intermediate goods leads to tougher local competition after a reduction of τ_1 .

¹²It follows $\kappa_2 > 0$ and $\frac{P_{X,1}}{P_{X,2}} > \frac{1}{e^{\tau_0}}$.

¹³This is because the direct effect through final goods only intensifies the competition between 1l and 1s.

¹⁴See Appendix for a detailed proof.

intermediate goods, (ii) have an uncertain effect on cutoff costs in domestic seaboard city 1s through cost-saving effect of intermediate goods, (iii) reduce the cutoff costs in all cities through the final-goods transportation effect.

2.3.2 Domestic Trade Cost and Cutoff Cost for Exporting

The cutoff costs for firms located in domestic cities 1l and 1s to export to the foreign market 2s are given by:¹⁵

$$c_{M,1s2s} = \frac{c_{M,2s2s}}{e^{\tau_0}} = \left(\frac{\gamma}{L_{2s}\zeta} \frac{\kappa_2 \rho_0^{(\xi+2)/\xi}}{(1 - \rho_0^2)(1 + \rho_2)} \right)^{\frac{1}{\xi+2}}, \quad (45)$$

$$c_{M,1l2s} = \frac{c_{M,2s2s}}{e^{\tau_0 + \tau_1}} = \left(\frac{\gamma}{L_{2s}\zeta} \frac{\kappa_2 (\rho_0 \rho_1)^{(\xi+2)/\xi}}{(1 - \rho_0^2)(1 + \rho_2)} \right)^{\frac{1}{\xi+2}}. \quad (46)$$

Because $c_{M,1s2s}/c_{M,1l2s} = e^{\tau_1} > 1$, firms in the inland city face tougher export competition relative to the seaboard city. Considering the effect of τ_1 on cutoff costs for exporting, we have:

$$\begin{aligned} \frac{\partial c_{M,1s2s}}{\partial \tau_1} &= \Delta a_1 c_{M,1s2s} > 0, \\ \frac{\partial c_{M,1l2s}}{\partial \tau_1} &= (\Delta - 1) a_1 c_{M,1l2s}, \end{aligned} \quad (47)$$

where $\Delta \equiv \frac{\xi}{\xi+2} \frac{\kappa_5}{\kappa_2} \in (0, 1)$.¹⁶

In city 1s, the decrease in domestic trade costs only intensifies export competition to 2s through the intermediate goods price channel. There is no final-goods transportation effect because the international shipping cost between the two countries have not changed. However, for the inland city 1l, local goods have greater price advantage in the exporting market to 2s with lower domestic trade costs, which may offset or reverse the increase in competition caused by the decline in intermediate goods prices. In fact, when the cutoff cost in 2s of is sufficiently inelastic with respect to τ_1 , the final goods competition effect

¹⁵We omit the discussion on exporting to 2l because the analysis is similar to 2s, with the only difference being the additional domestic trade costs within country 2.

¹⁶Since $\kappa_2 - \kappa_5 = (1 - \chi)\theta_2(1 + \rho_0^2\rho_2) > 0$ and $\frac{\xi}{\xi+2} \frac{\kappa_5}{\kappa_2} < \frac{\xi}{\xi+2} < 1$, we have $0 < \Delta < 1$.

is greater than the competition effect of intermediate goods,¹⁷ which makes it more likely for 1l's firms to enter the export market.

Based on the discussion above, we introduce the following lemma about export competition:¹⁸

Lemma 2. *A reduction in the domestic trade cost in country 1 will: (i) enhance export competition in domestic cities of both 1s and 1l through the competition effect of intermediate goods, (ii) mitigate export competition in inland city 1l through the final-goods transportation effect.*

2.3.3 Domestic Trade Cost and Extensive and Intensive Margins of Export

Now we discuss how domestic trade costs affect the firms' export performance on export scopes (extensive margin), measured by the number of export varieties, and on export revenues of a existing exported variety (intensive margin).

On the extensive margin, from Equation (19), we know that $M_{1f2s}(c_0)$, the number of varieties exported by a firm with cost c from city $1f$ ($f = s, l$) to $2s$, is the largest m that satisfies $c_0 \leq \omega^m c_{M,1f2s}$ given the cutoff cost $c_{M,1f2s}$. This condition can be rewritten as:

$$m \leq \frac{\ln c_{M,1f2s} - \ln c_0}{\ln \omega^{-1}}, \quad f = s, l, \quad (48)$$

Obviously, the right side $\frac{\ln(c_{M,1f2s}) - \ln c_0}{\ln \omega^{-1}} \equiv h_{1f2s}$ determines the maximum value of m such that $\partial M_{1f2s}(c_0)/\partial h \geq 0$. Since $\partial h/\partial c_0 < 0$ and $\partial h/\partial c_{M,1f2s} > 0$, there are confounding effects from a reduction in domestic trade costs on export scope. As stated in Lemma 2, it raises the cutoff costs for exporting to $2s$ through the final-goods transportation effect and lowers the cutoffs through the competition effect of intermediate goods. On

¹⁷From Equation (47), $\frac{\partial c_{M,1l2s}}{\partial \tau_1} < 0$ if $\Delta < 1$. And Equation (43) implies that $\frac{\partial c_{M,2s2s}}{\partial \tau_1} = \Delta a_1 c_{M,2s2s}$. Therefore, $\Delta = \frac{\varepsilon_{c_{M,2s2s}, \tau_1}}{a_1 \tau_1} < 1$, where $\varepsilon_{c_{M,2s2s}, \tau_1} \equiv \frac{\partial c_{M,2s2s}/\partial \tau_1}{c_{M,2s2s}/\tau_1}$, is equivalent to $\varepsilon_{c_{M,2s2s}, \tau_1} < a_1 \tau_1$.

¹⁸See Appendix for a detailed proof.

the other hand, cost-savings from access to lower-priced domestic intermediate goods help firms surpass the export thresholds in more varieties. We show in the Appendix that the cost-saving effect dominates the competition effect of intermediate goods for firms in both 1s and 1l, thereby expanding their export varieties.

The result is summarized in the following proposition:

Proposition 1. *After a reduction of domestic trade costs: (i) the cost-savings effect of intermediate goods dominates the competition effect of intermediate goods, which leads domestic firms to expand their scopes of export varieties, (ii) final-goods transportation effect increases export scopes of domestic firms in the inland city.*

Next, we consider the impact of domestic trade costs on the intensive margin of exports. As in the previous subsection, improved access to intermediate goods generates both cost-saving and competition effects, which are the central mechanisms of interest. For a firm with core unit cost c_0 at city 1f ($f = s, l$), we take derivatives of export revenues of the m -th variety $r_{1f2s}(m, c_0)$ with respect to τ_1 by holding the export scope $M_{1f2s}(c_0)$ constant. It can be proved that:¹⁹

$$\frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} \leq 0 \text{ if } \frac{v(m, c_0)}{c_{M,1s2s}} \geq \Delta^{1/2}, \quad (49)$$

$$\frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} \leq 0 \text{ if } \frac{v(m, c_0)}{c_{M,1l2s}} \geq \left(\frac{\Delta}{1 + \nu} \right)^{1/2}, \quad (50)$$

where $\nu = 1/a_1 > 0$. This implies that only the export revenues of varieties with higher costs $v(m, c_0) = \omega^{-m}c_0$ increase with the decline in domestic trade costs. Higher product-level costs arise from two sources of heterogeneity. At the firm level, firms with higher core unit costs c_0 generally have higher production costs across all their products. Within a firm, varieties farther from the core competence incur higher costs as ω^{-m} increases with m .

¹⁹See Appendix for a detailed proof.

The impact of domestic trade cost reduction on product-level export revenues can also be decomposed into three effects. As shown in the Appendix, the cost-saving effect of intermediate goods increases with the cost of the final goods exported by the firm. When the cost exceeds a certain threshold relative to the export cutoff, such cost savings will dominate the competition effect of intermediate goods, which leads to an increase in export revenues. Additionally, for inland firms, the final-goods transportation effect generates extra cost savings due to reduced export transportation costs, which offsets part of the competition effect and helps more varieties with moderate cost to increase their export sales.

Rewriting Equations (49) and (50) based on the expression for markup λ , we obtain:

$$\frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} \leq 0 \text{ if } \lambda_{1s2s}(m, c_0) \leq \frac{c_{M,2s2s}}{2} (1 - \Delta^{1/2}), \quad (51)$$

$$\frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} \leq 0 \text{ if } \lambda_{1l2s}(m, c_0) \leq \frac{c_{M,2s2s}}{2} \left[1 - \left(\frac{\Delta}{1 + \nu} \right)^{1/2} \right]. \quad (52)$$

Since λ decreases monotonically with both c_0 and m , lower product markups stem from either a higher core unit cost of the firm (a higher c_0) or from producing more peripheral products (a larger m). In the empirical analysis in the next section, due to constraints of data availability, we focus on examining the heterogeneity across firms by using the indicator of firm-level cost information.

It is worth noting that Equations (49)-(52) are derived under a given export scope. That is, they apply only to the existing exported varieties by exporting firms.

We summarize the findings in the following proposition:

Proposition 2. *After the reduction of domestic trade cost: (i) for domestic firms, the cost-saving effect of intermediate goods is bigger (smaller) than the competition effect of intermediate goods for their existing exporting varieties with lower (higher) markups, leading to an increase (decrease) in their product-level export revenues, (ii) the final-goods*

transportation effect tends to increase product-level export revenues of domestic inland firms.

Finally, the aggregated firm-level export revenues at city 1s and 1l are given by:

$$R_{1s2s}(c_0) = \sum_{m=0}^{M_{1s2s}(c)-1} r_{1s2s}(m, c_0) = \sum_{m=0}^{M_{1s2s}(c)-1} \frac{L_{2s}}{4\gamma} (c_{M,2s2s}^2 - (e^{\tau_0} \omega^{-m} c_0)^2), \quad (53)$$

$$R_{1l2s}(c_0) = \sum_{m=0}^{M_{1l2s}(c)-1} r_{1l2s}(m, c_0) = \sum_{m=0}^{M_{1l2s}(c)-1} \frac{L_{2s}}{4\gamma} (c_{M,2s2s}^2 - (e^{\tau_0+\tau_1} \omega^{-m} c_0)^2). \quad (54)$$

We derive the following corollary regarding the impact of domestic trade costs.

Corollary 1. *A reduction in domestic trade costs could raise the total export revenues of domestic firms if at least one of the following two conditions holds: (i) the number of export varieties increases sufficiently, (2) given the export scope, the total export growth of low-markup varieties is greater than the export reduction of high-markup varieties.*

The empirical analysis focuses on the implications of improved access to domestic intermediate inputs. Accordingly, our main explanatory variable is an upstream market access measure that captures firms' access to domestic suppliers. Since reductions in domestic trade costs also affect the transportation of final goods, we control for firms' distance to seaports in the empirical analysis to separate this conventional channel from the effects of intermediate-input access.

3 Empirical Design and Data

Guided by theory, we proceed to examine how the domestic trade integration brought about by the reduction in domestic trade costs affects firms' export performance.

3.1 Regression Model

The regression model is as follows

$$\begin{aligned} \ln(\text{export}_{ijct}) = & \beta_0 + \beta_1 \ln(\text{UMA}_{jct}) + \beta_2 \ln(\text{EUMA}_{jct}) + \beta_3 \ln(\text{dist_port}_{ct}) + \beta_4 X_{ct} \\ & + \varphi_i + \omega_t + u_{ijct} \end{aligned} \quad (55)$$

where export_{ijct} is the exports of firm i in industry j and city c in the year t . The dependent variable in the benchmark regression is the log of the firm-level export value. In further empirical analysis, we also use firms' export varieties, destination countries, variety-destination combinations, and firm-product-level export values as the dependent variables.

The variable UMA_{jct} represents the upstream market access (UMA) of industry j in city c . Extending the method of Donaldson and Hornbeck (2016), we construct the UMA index by combining geographical information from the railway network with inter-industry input-output relations. This index measures the spatial integration of firms with their upstream industries via the domestic transportation network, thereby depicting their access to domestic intermediate goods. The details on the UMA index are contained in Subsection 3.2.

To achieve identification of β_1 , the main parameter of interest, we include in the regression the expected level of MA for industry j and city c in year t , given the city's geographic location and level of economic development (EUMA_{jct}). Our identification strategy is discussed in subsection 3.3.

Our benchmark regressions additionally control for the railway distance from city c to its nearest seaport city²⁰, dist_port_{ct} , allowing us to distinguish the effects of domestic intermediate-input access from the more conventional channel of lower transportation costs for final goods.

²⁰It refers to the major ports engaged in international trade in mainland China; see Appendix C for the list of major ports published by the Ministry of Transport of China in 2004.

We include several control variables to capture the effects of both city-level and firm-level factors. X_{ct} is a vector of prefecture-level control variables that may affect firms' export behaviors. They are log GDP ($lngdp$), log GDP per capita ($lngdppc$), the ratio of GDP of the secondary to GDP of tertiary industries ($indserv$), and highway density measured as kilometers of highway per square kilometer ($highway$).²¹ The variables φ_i , ω_t , and u_{ijct} represent the firm fixed effect, year fixed effect, and the error term, respectively.

3.2 Construction of Upstream Market Access

Following the market access (MA) approach of Donaldson and Hornbeck (2016), we construct an UMA index based on connectivity to freight railway in China. The UMA index we construct differ from the MA index of Donaldson and Hornbeck (2016) in two ways. First, we incorporate inter-industry input-output information. Second, by using information on the annual average freight rate, our UMA index contains temporal changes in market access due to prices. Specifically, the upstream market access of industry j in city c at year t is given by:

$$UMA_{jct} = \sum_{c' \neq c} \tau_{cc',t}^\theta \sum_{j' \neq j} a_{jj'} \sum_{i' \in I(j',c')} VA_{i'j'c',t} \quad (56)$$

where $VA_{i'j'c',t}$ is the value added of firm i' of industry j' in domestic city c' at year t with $I(j',c')$ representing the set of firms of industry j' in city c' . Therefore, the sum $\sum_{i' \in I(j',c')} VA_{i'j'c',t}$ represents the aggregate value added at the city-industry level. The factor $a_{jj'}$ is the input-output coefficient, measured by the proportion of total output of industry j that comes from the intermediate inputs of industry j' at the national level. Therefore, j' is an upstream industry to j if $a_{jj'}$ is positive. The input-output coefficients are obtained from the 2002 Chinese Input-output Tables.

$\tau_{cc',t}^\theta$ is an index of railway transportation cost per ton of goods shipped between city c and c' at year t . Following Baum-Snow et al. (2017), the iceberg trade cost $\tau_{cc',t}$ is

²¹Due to the lack of historical road network data at the city level, we use province-level road density as a proxy.

assumed to be a function of the annual average freight rate (p_t) and the distance between two cities by railway ($d_{cc',t}$)

$$\tau_{cc',t} = 1 + p_t(d_{cc',t})^\sigma.$$

We obtain the average freight rate in a year, p_t , by dividing the total revenue of railway transportation deflated with production price index by the total freight volume in tons. The distance variable ($d_{cc',t}$) is the shortest railway distance between two cities which we calculate based on Dijkstra et al. (1959). The detailed steps for constructing the railway vector map are described in Appendix A. The parameter σ captures the concave relationship between shipping cost and railway distance, which is set to 0.8 (Baum-Snow et al., 2017). θ is the price elasticity of trade that measures the substitution of outputs between different cities. As in Donaldson (2018), we set the value of θ to 3.8.

Our UMA index captures differences among firms in access to domestic upstream intermediate goods in three dimensions. Firstly, in terms of spatial dimension, geographic remoteness hinders connectivity between local firms and those in other cities. The second aspect is the heterogeneity in industrial distribution. Among enterprises within the same city, if neighboring cities are densely populated with upstream firms related to a particular industry, then local firms in that industry will have better access to intermediate goods. The third aspect is time variation. Cities with rapid railway development will enable their firms in obtaining more intermediate goods from other cities at lower costs.

3.3 Identification Strategy

Identifying the causal effect of upstream market access (UMA) on firm exports is challenging because railway expansion is unlikely to be randomly assigned across locations. Railway investments may be correlated with local economic conditions, growth prospects, or policy objectives, all of which can also affect firms' export performance. As a result, OLS estimates may not recover the causal effect of improved access to domestic interme-

diate goods.

To address this concern, we follow Borusyak and Hull (2023) and combine an instrumental-variables approach with a control-function strategy. The key insight is that even after instrumenting for transportation connectivity, part of the variation in market access may still reflect persistent geographic advantages. We therefore include the expected level of upstream market access (EUMA) in the second-stage regression to absorb the component of UMA that is predictable from geography and long-run economic fundamentals. Identification is then based on deviations of realized UMA from its expected level.

To generate exogenous variation in UMA, we construct two artificial transportation networks linking 36 major cities in mainland China.²² Following Faber (2014), the first network is based on least-cost paths, while the second follows Banerjee, Duflo, and Qian (2020) and connects cities using straight lines. The artificial transportation networks are illustrated in Figure 2. The left and right panels correspond to the least-cost-path and straight-line networks, respectively. In each panel, the denser network shown by thinner lines represents the actual railway network in 2007. The close correspondence between the artificial and actual networks suggests that the former capture important determinants of realized railway connectivity.

Using shortest-path distances implied by the artificial networks and the upstream-downstream input-output linkages weighted by base-year industry value added, we construct two counterfactual UMA measures. Since these measures are time invariant, we interact them with the annual growth rate of nationwide fixed-asset investment to obtain time-varying instruments for realized UMA.²³ We also construct instruments for railway distance to the nearest seaport using distances implied by the artificial networks. This

²²The major cities include the 27 provincial capitals in Mainland China, four municipalities directly under the Central Government (Beijing, Shanghai, Tianjin, and Chongqing), and five additional vice-provincial cities that are not provincial capitals (Dalian, Ningbo, Qingdao, Shenzhen, and Xiamen).

²³We remove the long-run trend in the investment growth series using the Hodrick-Prescott filter.

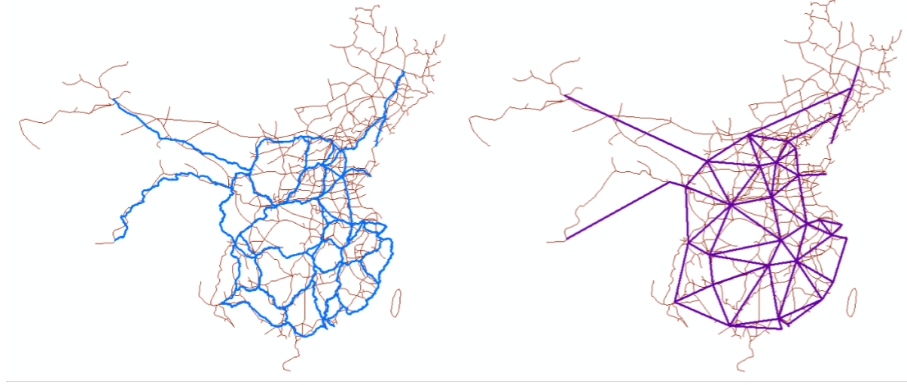


Figure 2: Artificial Transportation Networks Predicted by Exogenous Factors

Source: authors' calculations. Note: The blue network in the left panel and the purple network in the right panel are the artificial transportation networks.

allows us to separately identify the effects of access to domestic intermediate inputs (captured by UMA) and transportation costs for final goods (captured by distance to the nearest seaport).

In addition, our benchmark specification includes the railway distance from each prefecture to its nearest seaport. We construct corresponding instruments using distances implied by the artificial transportation networks. Specifically, the counterfactual distance to the nearest seaport, interacted with the nationwide fixed-asset investment growth rate, serves as an instrument for the actual railway distance to the nearest seaport.

Under this framework, the coefficient on UMA captures the causal effect of exogenous changes in access to domestic intermediate goods, while EUMA controls for the component of market access attributable to persistent geographic and historical factors.

3.4 Data

Our regressions are merged data from two firm-level datasets, the firm-level data from the Annual Survey of Industrial Firms (ASIF) and trade transaction data from Chinese Customs Database (CCD) from 2000 to 2007.

The ASIF firm-level data. The data are collected by the National Bureau of Statis-

tics (NBS) of China in an annual surveys of manufacturing enterprises, covering all Chinese state-owned enterprises (SOEs) and non-SOEs with sales greater than five million yuan (\$700,000 at current exchange rate). We follow the steps of Brandt, Van Biesebroeck, and Zhang (2012), Feenstra, Li, and Yu (2014) and Yu (2015) to remove unreliable observations from the sample. Specifically, we delete the observations with missing key financial variables (including total assets, fixed assets, number of employees, total sales, value of gross outputs and value added) and observation that report contradictory financial variables: (i) total fixed assets are greater than total assets; (ii) liquid assets are greater than total assets; (iii) the net value of fixed assets is greater than total assets; (iv) the depreciation for the current year exceeds the accumulated depreciation; and (v) any of the variables above is less than zero.

Transaction-level trade data. The Chinese Customs Database provides detailed information on China’s trade transactions of both imports and exports. The dataset provides important variables for each transaction, such as trading price, quantity and value, product code at the HS eight-digit level, source or destination country. Most importantly, it records the identity information of the enterprises engaged in trade, allowing researchers to merge it with the dataset of firm-level production data.

Merged dataset. We first match the production data and trade transaction data following the steps of Yu (2015) by using firms’ names, zip codes, as well as phone numbers. Then they are merged by the datasets of upstream market access index we construct. To establish a one-to-one correspondence between the firms and their upstream industries, we convert the 4-digit Chinese industry classification (CIC) industry codes in the firm-level data to the industry classification in the 2002 Input-output Table with 128 industries.²⁴ Then we only retain the firms in the manufacturing sector as my interest is the relation of domestic intermediate inputs and firms’ export behaviors. We also merge to the sample

²⁴As the CIC implemented standard revisions in 2003, we first follow Brandt, Van Biesebroeck, and Zhang (2012) to unify the pre- and post-revision industry classifications into an adjusted industry code.

city-level characteristic variables, which are obtained from the China City Statistical Yearbook. After the steps, we obtain an unbalanced panel of 271,975 firm-year observations, covering 94,201 export firms during 2000-2007. Summary of key variables can be found in Table 3.1.

Figure 3 depicts the trends of mean firm-level exports over time. The average export value of firms experienced continuous stable growth from 2000 to 2006, followed by a decline in 2007. Meanwhile, the average export scope, i.e., the number of export varieties, also increased by approximately 50% during this period. However, at the intensive margin, the average export value per variety remained almost unchanged or even experienced a slight decrease. This indicates that China's export growth during this period was primarily driven by the expansion of the extensive margin.

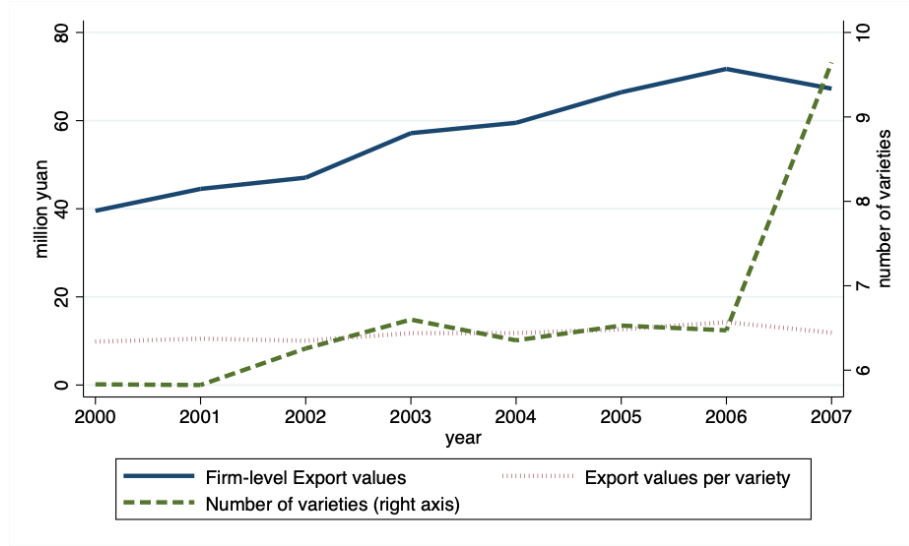


Figure 3: Firm-level exports
Source: authors' calculations.

As per the model's theoretical setup, we divide the locations of firms into seaboard and inland cities. Specifically, the seaboard region comprises eight coastal provinces (Liaoning, Shandong, Jiangsu, Zhejiang, Fujian, Guangdong, Guangxi, Hainan) and two

province-level municipalities (Tianjin and Shanghai), while the inland region consists of the remaining provinces of mainland China. Table 3.2 presents a detailed comparison of export performance between seaboard and inland firms across different dimensions: the number of exporters, average exports, export scope (number of varieties), export destinations (number of countries) and cost-to-sales ratios at the firm level. Consistent with Lemma 3.1 in the model, the seaboard firms outperform inland firms in exports. On average, seaboard firms have higher export revenues (61.38 compared to 59.82 million yuan) and export a broader scope of products (7.13 varieties vs. 7.01 varieties) and to more countries (8.54 countries vs. 7.34 countries). There are 74,096 seaboard exporters throughout the sample period, which is 3.7 times the number of inland (with 20,105 export firms). In addition, the average cost-to-sale ratio of seaboard export firms is higher than that of inland exporters (0.856 to 0.817), indicating a more stringent cutoff cost for entering the export market in the inland cities. All these facts are consistent with the model predictions. The t-values, however, indicate that disparities in export revenues and number of varieties are insignificant in the statistical sense.

4 Regression Results

4.1 Benchmark Export Results

Table 3 reports the OLS estimates for the benchmark specification. Column (1) presents a parsimonious specification including only UMA and EUMA. Consistent with our theoretical predictions, the coefficient on UMA is positive, indicating that improved access to domestic intermediate inputs is associated with higher firm exports. Quantitatively, a one-percent increase in UMA is associated with a 0.026% increase in exports, although the estimate is not statistically significant. Column (2) additionally controls for the railway distance to the nearest seaport city. The estimated coefficient on this variable is -0.027, suggesting that firms located closer to seaports export more. The coefficient on UMA

remains virtually unchanged at 0.025. Column (3) reports our preferred specification, which further incorporates city-level controls and an indicator for state-owned enterprises (SOEs). The estimated effect of UMA declines slightly to 0.022 but remains stable in magnitude, while the coefficient on railway distance becomes larger in absolute value (-0.071). Overall, the OLS estimates suggest a positive relationship between upstream market access and firm export performance.

To address the endogenous placement of railway routes, we follow Borusyak and Hull (2023) and combine an instrumental-variables approach based on counterfactual railway networks with a control-function strategy implemented through the inclusion of EUMA. As discussed in Subsection 3.3, we construct two counterfactual upstream market access measures as instruments. The first instrument, $UMA_{geocost}$, is based on an artificial railway network formed by least-cost paths, while the second, $UMA_{straight}$, is derived from an artificial network connecting cities by straight lines. We also construct two corresponding instruments for the railway distance to the nearest seaport using distances implied by these counterfactual networks.²⁵ For convenience, we refer to the set of instruments based on the least-cost-path network as IV1 and those based on the straight-line network as IV2.

Table 4 reports the estimates from the benchmark specification under our IV-control-function strategy. Using IV1 and IV2 as instruments in Columns (1) and (2), respectively, we find that a one-percent increase in upstream market access raises firm exports by 0.083% and 0.056%. These estimates are substantially larger than the corresponding OLS estimate reported in Column (3) of Table 3, suggesting that OLS understates the positive effect of upstream market access on firm exports. The estimated coefficient on the distance to the nearest seaport also increases in absolute value, although its statistical significance is sensitive to the choice of instrument. The Kleibergen–Paap rk LM statistics reject the null hypothesis of underidentification, while the Kleibergen–Paap rk Wald F statistics indicate

²⁵Following the construction of the UMA instruments, these distance instruments incorporate temporal variation through interaction with the national growth rate of fixed-asset investment.

that weak instruments are unlikely to be a concern.²⁶ In unreported estimates, we also use IV1 and IV2 simultaneously as instruments; the results are very similar to those reported in Table 4. However, following the recommendation of Angrist and Kolesár (2024), we focus on just-identified IV estimators. Because IV1 exhibits substantially stronger first-stage performance, we use it as our preferred instrument set in the subsequent analysis.

4.2 The Dual Margin of Exports

Having established the positive effect of upstream market access on firm-level exports in Table 4, we now examine the mechanisms underlying this effect by testing the predictions of Propositions 1 and 2. Specifically, we investigate whether improved access to domestic intermediate inputs affects exports through both the extensive and intensive margins. Because the extensive margin is not uniquely defined in the literature, we consider several commonly used measures. Following Hummels and Klenow (2005), Amiti and Freund (2008), and Shepherd (2010), we measure the extensive margin by the number of export varieties. We also examine the number of destination countries, as in Helpman, Melitz, and Rubinstein (2008) and Felbermayr and Kohler (2006), and the number of variety-destination combinations, following Evenett (2002). According to Proposition 1, improved upstream market access should increase each of these measures of export scope.

Table 5 reports the effects of upstream market access on firms' export scope. Guided by our model, our primary measure of the extensive margin is the number of export varieties. The estimates in Columns (1) and (2) indicate that a 1% increase in UMA raises the number of export varieties by 0.032% and 0.024% at the HS6 and HS4 levels, respectively. In Column (3), we use the number of export destinations as an alternative measure of the extensive margin and again find a positive and statistically significant effect of UMA. Columns (4) and (5) further examine the number of destination–variety combinations at

²⁶Appendix C reports the corresponding first-stage results. As an additional robustness check, we also estimate the specification with industry-year fixed effects using 2SLS. These results are likewise reported in Appendix C.

the HS6 and HS4 levels. The estimated coefficients on UMA remain positive and significant for both measures. Overall, the results consistently support Proposition 1, which predicts that improved access to domestic intermediate inputs expands firms' export scope.

Comparing the coefficients on seaport distance across specifications also provides suggestive evidence on the role of final-goods transportation costs. The estimated effects of seaport distance are generally stronger for export destinations and destination–variety combinations than for export varieties alone, suggesting that lower trade costs to seaports primarily facilitate firms' access to foreign markets rather than the expansion of their product range.

Table 6 examines the effect of UMA on the intensive margin of exports. Following the alternative definitions of export scope used in Table 5, we construct corresponding measures of the intensive margin by dividing a firm's total exports by the number of export varieties, export destinations, and destination–variety combinations, respectively. The estimates in Columns (1) and (2) show that a 1% increase in UMA raises average exports per variety by 0.044% and 0.054%, with the coefficients statistically significant at the 10% and 5% levels, respectively. By contrast, the estimated coefficients in Columns (3)–(5) are 0.015, 0.020, and 0.019, and none is statistically significant at conventional levels. The coefficients on seaport distance exhibit a similar pattern. These results suggest that lower trade costs primarily increase exports at the product level rather than exports per destination or destination–variety combination.

Taken together with the results in Table 5, the evidence indicates that improved access to domestic intermediate inputs robustly expands firms' export scope and may also increase exports along certain dimensions of the intensive margin. We next turn to a more direct test of Proposition 2 by examining the heterogeneous effects of UMA on product-level exports.

Table 7 implements this test by examining the heterogeneous effects of upstream

market access on product-level exports. Proposition 2 predicts that improved access to intermediate goods increases product-level exports for low-markup varieties but reduces them for high-markup varieties, as the cost-saving effect of intermediate goods dominates the competition effect of intermediate goods only for the former. To test this prediction, we regress firm-product-level exports at the HS6 level on upstream market access. Unlike Table 6, which examines exports averaged across a firm’s export scope, the unit of observation here is a firm-product pair, matching the level at which Proposition 2 is derived. We additionally control for HS6 product fixed effects to account for product-specific characteristics. The estimate in Column (1) shows that UMA reduces product-level exports on average, consistent with the dominance of the competition effect of intermediate goods for a large share of products.

Because product-level markups are not observed, we test Proposition 2 using firm-level cost ratios, defined as total costs divided by revenue, as an inverse measure of markup. A higher cost ratio corresponds to a lower markup and, according to the model, products of such firms should benefit more from improved access to intermediate inputs.²⁷ Column (2) introduces the interaction between UMA and the cost ratio. The positive and statistically significant interaction coefficient supports the prediction of Proposition 2: firms with higher costs (lower markups) experience more favorable product-level export responses to improvements in UMA.²⁸

Columns (3) and (4) provide complementary evidence using indicator variables for firms above the 50th and 75th percentiles of the cost-ratio distribution. Relative to low-cost firms, the UMA elasticity of product-level exports is 0.109 higher for firms above the median cost ratio. When the threshold is raised to the 75th percentile, the estimated interaction remains positive but becomes smaller and statistically insignificant.²⁹ Across

²⁷In the model, higher-cost firms and varieties are more likely to experience export gains because the cost-saving effect of intermediate goods dominates the competition effect.

²⁸The coefficient on $\ln(UMA)$ itself should be interpreted with caution, as it corresponds to firms with cost ratios approaching zero.

²⁹Appendix C reports analogous regressions at the HS4 level, yielding similar results.

all specifications, the coefficient on distance to the nearest seaport remains negative, with magnitudes ranging from -0.057 to -0.055, although the estimates are not statistically significant.

Overall, the evidence in Tables 5 to 7 is broadly consistent with the model’s predictions. Improved access to domestic intermediate inputs robustly expands firms’ export scope and, for some measures, increases the intensive margin. More importantly, firms with higher cost ratios benefit disproportionately from improvements in upstream market access, providing direct support for Proposition 2.

4.3 Seaboard vs. Inland

In Panel A of Table 8, we estimate the baseline 2SLS regression for firms in seaboard provinces. The coefficient on UMA (0.101) is larger than that of the benchmark regression in Table 4 (0.083). Further, we examine the effects on extensive margins of exports in Columns (2)-(4), where the dependent variables are the number of HS6-level export varieties, HS4-level varieties, and export destinations. The coefficients are comparable to or larger than the corresponding ones in Table 5. Overall, the UMA elasticity of exports appears to be larger for firms in seaboard provinces. In all columns, the exports of seaboard firms decrease with the distance to ports. This is because firms in the seaboard provinces are not all located in port cities, and therefore their export transportation costs can be reduced by railway construction.

As for firms in inland provinces, estimates in Panel B indicate that UMA has no significant effects on exports, except for increasing the number of export destinations. The effects of distance to ports are also insignificant in each of the regressions. Therefore, most of the export benefits associated with better access to intermediate goods and lower costs of shipping final goods are concentrated among firms in seaboard provinces.

To understand the effect of UMA on intensive margins, we replicate the intensive margin regressions in the last three columns of Table 7 with subsamples of seaboard firms

and inland firms. In general, the coefficients on UMA and its interaction terms based on seaboard firms (the first through third columns of Table 9) and inland firms (the fourth through sixth columns of Table 9) are similar, in terms of signs and magnitude, to those of the full sample. However, they often lack statistical significance.

Overall, the heterogeneity results are consistent with the mechanisms highlighted in the model. For firms in seaboard provinces, improved access to domestic intermediate inputs lowers production costs and facilitates access to foreign markets, leading to sizable gains in export performance. For inland firms, however, the cost-saving effect is partially offset by stronger competition from firms located closer to ports. Consequently, the positive effects of upstream market access on exports are concentrated among seaboard firms, whereas the estimated effects for inland firms are generally small and statistically insignificant.

4.4 Evidence on the Cost-Saving and Competition Channels

We first examine the cost-saving channel by estimating the effect of UMA on several measures of production costs. Column (1) of Table 10 uses the logarithm of intermediate-input expenditure as the dependent variable and shows that a 1% increase in UMA reduces input costs by 0.038%. To account for differences in firm size, Column (2) uses the share of intermediate inputs in total costs as the dependent variable. The estimate implies that a 1% increase in UMA lowers this share by 0.011 percentage points. Column (3) uses the ratio of intermediate-input costs to revenue and finds a reduction of 0.007 percentage points. Finally, Column (4) shows that a 1% increase in UMA is associated with a 0.022% decline in total costs, although the estimate is not statistically significant at conventional levels. Overall, these results suggest that improved access to domestic intermediate inputs reduces firms' production costs.

We next examine the competition channel. Because the entry and exit of non-exporting firms may also affect market competition, we use the full manufacturing sample

rather than the matched customs sample. Firm outcomes are aggregated to the city-industry level to construct the fraction of exporters, the entry rate, the exit rate, and the net entry rate. Column (1) of Table 11 shows that a 1% increase in UMA raises the share of exporting firms by 0.143%. Columns (2) and (3) indicate that UMA increases entry and reduces exit, although only the latter effect is statistically significant. As a result, Column (4) shows that a 1% increase in UMA raises the net entry rate by 0.37%. These results suggest that improved access to intermediate inputs attracts additional firms into local markets and increases the density of competitors.

Taken together, the results in Table 10 and Table 11 provide empirical support for the two mechanisms emphasized in the model. On the one hand, improved upstream market access lowers firms' production costs, consistent with the cost-saving effect of better access to domestic intermediate inputs. On the other hand, higher UMA increases net firm entry and the share of exporters, indicating more intense local competition. These findings suggest that reductions in domestic trade costs simultaneously improve firms' access to cheaper intermediate goods and reshape the competitive environment in local markets. The coexistence of these two forces helps explain the heterogeneous effects of UMA on export performance documented in the previous subsection.

4.5 Robustness Checks

Table 12 examines whether the effect of upstream market access persists over time. Column (1) reproduces the benchmark specification from Table 4, while Columns (2)–(4) replace contemporaneous UMA with its one-, two-, and three-year lags, respectively. The estimated effect of upstream market access declines rapidly with the lag length, as both the magnitude and statistical significance of the coefficients diminish substantially. This pattern suggests that firms respond promptly to improvements in upstream market access, with most of the export adjustment occurring contemporaneously rather than with a substantial lag. One possible explanation is that railway expansions are anticipated,

allowing firms to adjust production and export decisions in advance.

Table 13 presents a series of additional robustness checks. First, to ensure that the benchmark results are not driven by firms located in major cities, Column (1) of Table 13 excludes firms in the 36 major Chinese cities, including the four municipalities directly under the central government, 27 provincial capitals, and five additional major cities described in Section 3.3. Second, Column (2) restricts the sample to a balanced panel of exporters that report positive export values throughout 2000–2007, thereby alleviating concerns that compositional changes among exporting firms drive the results. Third, Column (3) controls for economic agglomeration by including a prefecture-level clustering index following Ruan and Zhang (2025). The index incorporates both market size and industry relatedness and is constructed using annual firm asset values from ASIF as weights. Fourth, Column (4) employs an alternative measure of upstream market access, $UMA^{total\ output}$, which uses total output rather than value added in its construction. Across all specifications, the estimated effect of UMA remains positive, statistically significant, and quantitatively similar to the benchmark estimate.

5 Conclusion

This paper examines how domestic trade integration, driven by transportation infrastructure improvements, affects the export performance of Chinese manufacturing firms through access to domestic intermediate goods.

To study this question, we extend the multi-product firm model of Mayer, Melitz, and Ottaviano (2014) by incorporating intermediate-input trade. The model highlights two opposing forces generated by reductions in domestic trade costs: a cost-saving effect arising from improved access to intermediate goods and a competition effect resulting from intensified market competition. We show that the cost-saving effect dominates at the extensive margin, leading firms to expand their export product scope. Along the

intensive margin, however, the effects on product-level exports are heterogeneous. Lower-markup varieties benefit more from improved access to intermediate inputs, while higher-markup varieties are more exposed to intensified competition. As a result, firm-level exports increase when the expansion of export varieties and the growth of exports among lower-markup products outweigh the negative effects on higher-markup products.

We test these predictions using firm-level export data for Chinese manufacturing firms during 2000–2007 and a novel upstream market access (UMA) index that combines information on railway network expansion with inter-industry input-output linkages. After addressing the endogeneity of railway placement through an instrumental-variables and control-function strategy, we find robust evidence that improved upstream market access increases firm exports. This effect operates primarily through the expansion of export scope, including export varieties, destinations, and destination–variety combinations. Consistent with the model, the effects on product-level exports are heterogeneous, with firms characterized by lower markups benefiting more from improvements in upstream market access. We further provide evidence for the two mechanisms emphasized in the model: improved access to domestic intermediate inputs reduces production costs while simultaneously intensifying local market competition through increased firm entry.

Taken together, our findings highlight the importance of domestic intermediate-input access as a determinant of export performance. Beyond reducing transportation costs for final goods, transportation infrastructure promotes exports by improving firms’ integration with domestic production networks. These results suggest that the benefits of infrastructure investment extend beyond market connectivity and include gains from more efficient sourcing of intermediate inputs within the domestic economy.

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Table 1: Summary Statistics

Variables	Unit	Obs.	Mean	Min	Max	S.D.
<i>export</i>	million yuan	306058	59.57	0	176117.3	742.16
<i>ln(UMA)</i>	\	40620	15.47	4.58	17.65	0.67
<i>ln(EUMA)</i>	\	40620	18.7	9.44	38.69	2.49
<i>ln(UMA)_geocost</i>	\	40620	15.23	11.1	17.49	0.68
<i>ln(UMA)_straight</i>	\	40620	15.27	11.41	17.59	0.64
<i>dist_port</i>	km	2385	732.72	0	6947.65	792.82
<i>GDP</i>	billion yuan	2385	5.56	0.19	121.89	8.45
<i>GDPpercapita</i>	yuan	2385	14419.33	1475.92	320254.8	18937.54
<i>industry/service</i>	\	2385	1.38	0.07	10.55	0.84
<i>highwaydensity</i>	km/km ²	247	7.69	0.07	104.84	11.46
<i>SOE</i>	binary	306058	0.05	0	1	0.23

Note: *ln(UMA)* is the log of upstream market access index, *ln(EUMA)* is the log of expected *UMA*, *ln(UMA)_geocost* is the log of counterfactual *UMA* that is constructed by the artificial transportation network connected by the least costly paths, and *ln(UMA)_straight* is the log of counterfactual *UMA* that is constructed by the artificial transportation network connected by straight lines. *dist_port* is the distance to the nearest port city. *GDP* and *GDPpercapita* are the GDP and GDP per capita at the prefecture level, and *industry/service* is the ratio of GDP of the secondary industries to tertiary industries. *highwaydensity* is province-level highway density measured by highway length per area. *SOE* is an indicator variable for state-owned enterprises.

Table 2: Comparison of Seaboard and Inland Firms's Exports

Statistics	Unit	Seaboard	Inland	Diff.	t-value
observations	count	202452	58192	98846	\
number of exporters	count	65372	38856	26516	\
average exports	million yuan	64.47	49.98	14.49	5.94
export scope	number of varieties	7.63	6.28	1.34	21.09
export destination	number of countries	8.58	7.59	0.99	26.44

Note: Columns (1) and (2) represent the means of export performance of seaboard and inland firms. Column (3) calculates the differences between the first two columns with values of the t-statistic displayed in Column (4).

Table 3: Upstream Market Access and Firm Exports (OLS)

Dependent Variables:	$ln(export)$		
	(1)	(2)	(3)
$ln(UMA)$	0.026 (0.029)	0.025 (0.029)	0.022 (0.029)
$ln(EUMA)$	-0.001 (0.008)	-0.001 (0.008)	-0.003 (0.008)
$ln(dist_port)$		-0.027 (0.037)	-0.071* (0.043)
$ln(GDP)$			-0.245*** (0.051)
$ln(GDP\ per\ capita)$			0.062 (0.042)
$secondary/tertiary$			-0.071* (0.043)
$ln(highway\ density)$			0.155*** (0.041)
SOE			-0.002 (0.038)
Dependent Variable's Mean	13.907	13.907	13.907
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
R-squared	0.805	0.805	0.805
Observations	268,168	268,168	268,168

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of firm exports. $ln(UMA)$ and $ln(EUMA)$ is the log of upstream market access index and the log of expected upstream market access. $ln(dist_port)$ is the log of distance to the nearest ports. $ln(GDP)$ and $ln(GDP\ per\ capita)$ are the logs of GDP and GDP per capita. $secondarytotertiary$ is the GDP ratio of the secondary to tertiary industries. $ln(highway\ density)$ is the log of province-level highway density measured by highway length per area. SOE is an indicator variable for state-owned enterprises.

Table 4: Upstream Market Access and Firm Exports (2SLS, Second Stage)

Dependent Variables:	$\ln(\text{export})$	
	IV1 (1)	IV2 (2)
$\ln(UMA)$	0.083*** (0.029)	0.056* (0.029)
$\ln(EUMA)$	-0.005 (0.008)	-0.004 (0.008)
$\ln(\text{dist_port})$	-0.084 (0.053)	-0.124** (0.055)
Dependent Variable's Mean	13.907	13.907
Time-varying firm controls	Yes	Yes
Time-varying city controls	Yes	Yes
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
K-P rk LM statistic	70.242	63.130
K-P rk Wald F statistic	1347.183	83.793
Observations	268,168	268,168

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of firm exports. The coefficients for Column (1) are estimated using IV1 which constructed by the artificial transportation network connected by the least costly paths. The coefficients for Column (2) are estimated using IV2 which is constructed by the artificial transportation network connected by connected by straight lines. Column (3) uses all IVs from the first two columns and conducts an overidentification test. $\ln(UMA)$ and $\ln(EUMA)$ is the log of upstream market access index and the log of expected upstream market access.

Table 5: Upstream Market Access and Extensive Margins of Export (2SLS)

Dependent Variables:	$\ln M^{hs6}$	$\ln M^{hs4}$	$\ln D$	$\ln(M^{hs6} \times D)$	$\ln(M^{hs4} \times D)$
	(1)	(2)	(3)	(4)	(5)
$\ln(UMA)$	0.032*** (0.010)	0.024*** (0.009)	0.058*** (0.013)	0.057*** (0.013)	0.057*** (0.014)
$\ln(EUMA)$	-0.008*** (0.003)	-0.006*** (0.002)	-0.001 (0.004)	-0.004 (0.004)	-0.003 (0.004)
$\ln(dist_port)$	-0.002 (0.013)	-0.001 (0.010)	-0.030 (0.018)	-0.031 (0.019)	-0.033* (0.019)
Dependent Variable's Mean	1.642	1.407	1.834	2.286	2.210
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	70.242	70.242	70.242	70.242	70.242
K-P rk Wald F statistic	1347.183	1347.183	1347.183	1347.183	1347.183
Observations	268,168	268,168	268,168	268,168	268,168

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variables in Columns (1) and (2) are the logs of number of export varieties at the HS6 and HS4 level, respectively. The dependent variable in Column (3) is the log of number of exporting destination countries. The dependent variables in Columns (4) and (5) are the logs of number of combinations of exporting destinations and export varieties at the HS6 and HS4 level, respectively.

Table 6: Upstream Market Access and Intensive Margins of Export (2SLS)

Dependent Variables:	$\ln \frac{export}{M^{hs6}}$	$\ln \frac{export}{M^{hs4}}$	$\ln \frac{export}{D}$	$\ln \frac{export}{M^{hs6} \times D}$	$\ln \frac{export}{M^{hs4} \times D}$
	(1)	(2)	(3)	(4)	(5)
$\ln(UMA)$	0.044*	0.054**	0.015	0.020	0.019
	(0.026)	(0.027)	(0.031)	(0.025)	(0.026)
$\ln(EUMA)$	0.005	0.003	-0.004	-0.000	-0.002
	(0.007)	(0.007)	(0.009)	(0.007)	(0.007)
$\ln(dist_port)$	-0.083*	-0.082*	-0.055	-0.051	-0.049
	(0.046)	(0.048)	(0.044)	(0.040)	(0.041)
Dependent Variable's Mean	12.579	12.874	12.352	11.814	11.814
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	70.242	70.242	70.242	70.242	70.242
K-P rk Wald F statistic	1347.183	1347.183	1347.183	1347.183	1347.183
Observations	268,168	268,168	268,168	268,168	268,168

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variables in Columns (1) and (2) are the logs of exports per export varieties at the HS6 and HS4 level, respectively. The dependent variable in Column (3) is the log exports per exporting destination country. The dependent variables in Columns (4) and (5) are the logs of exports per combination of exporting destinations and export varieties at the HS6 and HS4 level, respectively.

Table 7: Upstream Market Access and Product-level Exports

Dependent Variables:	$\ln(\text{export}^{hs6})$			
	(1)	(2)	(3)	(4)
$\ln(UMA)$	-0.009 (0.031)	-0.007 (0.031)	-0.062 (0.040)	-0.020 (0.034)
$\ln(EUMA)$	0.018** (0.007)	0.019*** (0.007)	0.018*** (0.007)	0.018** (0.007)
$\ln(\text{dist_port})$	-0.057 (0.046)	-0.056 (0.046)	-0.055 (0.046)	-0.057 (0.046)
$\ln(UMA) \times \text{costr}$		0.007*** (0.002)		
$\ln(UMA) \times \mathbb{1}(\text{costr}_{50th})$			0.109* (0.056)	
$\ln(UMA) \times \mathbb{1}(\text{costr}_{75th})$				0.046 (0.064)
Dependent Variable's Mean	11.454	11.454	11.454	11.454
Time-varying firm controls	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	38.852	39.044	38.892	39.324
K-P rk Wald F statistic	1690.357	1132.443	1127.867	1138.900
Observations	656,365	656,365	656,365	656,365

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of exports at the HS6-product level. *costr* is firm's cost rate measured by the total costs per unit of revenue. $\mathbb{1}(\text{costr}_{50th})$ is an indicator variable for firms with cost rates higher than the 50th percentile, and $\mathbb{1}(\text{costr}_{75th})$ is an indicator those with cost rates higher than the 75th percentile.

Table 8: Upstream Market Access and Exports of Firm from Different Regions

Dependent Variables:	$\ln(\text{export})$	$\ln M^{hs6}$	$\ln M^{hs4}$	$\ln D$
	(1)	(2)	(3)	(4)
Panel A. Seaboard				
$\ln(UMA)$	0.101*** (0.036)	0.043*** (0.013)	0.035*** (0.011)	0.058*** (0.017)
$\ln(EUMA)$	-0.015 (0.011)	-0.013*** (0.003)	-0.011*** (0.003)	-0.003 (0.006)
$\ln(\text{dist_port})$	-0.145** (0.063)	-0.004 (0.019)	-0.008 (0.014)	-0.071** (0.034)
Dependent Variable's Mean	14.044	1.700	1.449	1.854
K-P rk LM statistic	32.940	32.940	32.940	32.940
K-P rk Wald F statistic	333.077	333.077	333.077	333.077
Observations	180,492	180,492	180,492	180,492
Panel B. Inland				
$\ln(UMA)$	0.050 (0.043)	0.003 (0.016)	-0.005 (0.015)	0.054*** (0.019)
$\ln(EUMA)$	-0.002 (0.009)	0.002 (0.003)	0.002 (0.003)	0.003 (0.004)
$\ln(\text{dist_port})$	-0.056 (0.093)	0.006 (0.022)	0.016 (0.016)	0.002 (0.025)
Dependent Variable's Mean	13.625	1.522	1.321	1.791
K-P rk LM statistic	33.854	33.854	33.854	33.854
K-P rk Wald F statistic	4288.601	4288.601	4288.601	4288.601
Observations	87,655	87,655	87,655	87,655
Time-varying firm controls	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable in Column (1) is the log of exports at the firm level. The dependent variables in Columns (2) and (3) are the logs of number of export varieties at the HS6 and HS4 level, respectively. The dependent variable in Column (4) is the log of number of exporting destination countries. Panel A presents the estimation results for firms located in seaboard cities and Panel B shows results for firms in inland cities.

Table 9: Upstream Market Access and Product-level Exports at Different Regions

Dependent Variables:	$ln(export^{hs6})$					
	Seaboard			Inland		
	(1)	(2)	(3)	(4)	(5)	(6)
$ln(UMA)$	0.003 (0.034)	-0.052 (0.047)	-0.008 (0.038)	-0.044 (0.066)	-0.094 (0.074)	-0.072 (0.073)
$ln(EUMA)$	0.014 (0.009)	0.013 (0.009)	0.012 (0.009)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)
$ln(dist_port)$	0.068 (0.067)	0.069 (0.067)	0.067 (0.067)	-0.204*** (0.075)	-0.203*** (0.075)	-0.204*** (0.075)
$ln(UMA) \times costr$	0.007*** (0.003)			0.006 (0.004)		
$ln(UMA) \times \mathbb{1}(costr_{50th})$		0.107 (0.066)			0.100 (0.106)	
$ln(UMA) \times \mathbb{1}(costr_{75th})$			0.041 (0.074)			0.093 (0.128)
Dependent Variable's Mean	11.425	11.425	11.425	11.535	11.535	11.535
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	25.024	24.918	25.199	7.028	7.029	7.029
K-P rk Wald F statistic	494.929	492.990	495.937	487.740	487.985	486.256
Observations	483,269	483,269	483,269	172,650	172,650	172,650

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of exports at the HS6-product level. $costr$ is firm's cost rate measured by the total costs per unit of revenue. $\mathbb{1}(costr_{50th})$ is an indicator variable for firms with cost rates higher than the 50th percentile, and $\mathbb{1}(costr_{75th})$ is an indicator for firms with cost rates higher than the 75th percentile. Columns (1)-(3) presents the estimation results for firms located in seaboard cities and Columns (4)-(6) are for firms in inland cities.

Table 10: Upstream Market Access and Cost Reduction

Dependent Variables:	$\ln(input\ cost)$	$\frac{input\ cost}{total\ cost}$	$\frac{input\ cost}{revenue}$	$\ln(total\ cost)$
	(1)	(2)	(3)	(4)
$\ln(UMA)$	-0.038** (0.016)	-0.011** (0.005)	-0.007* (0.003)	-0.022 (0.014)
$\ln(EUMA)$	0.006 (0.005)	0.004*** (0.001)	0.001 (0.001)	-0.000 (0.005)
$\ln(dist_port)$	-0.033 (0.021)	0.013** (0.007)	0.004 (0.004)	-0.052** (0.024)
Dependent Variable's Mean	10.394	0.927	0.776	10.503
Time-varying firm controls	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	70.242	70.242	70.242	70.242
K-P rk Wald F statistic	1347.183	1347.183	1347.183	1347.183
Observations	268,168	268,168	268,168	268,168

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable in Column (1) is the log of input costs. The dependent variables in Columns (2) and (3) are input costs per unit of total costs and total revenue, respectively. The dependent variable in Column (4) is the log of total costs.

Table 11: Upstream Market Access and Market Competition

Dependent Variables:	$\frac{N_{export}}{N_{city \times ind}}$	$\frac{N_{entry}}{N_{city \times ind}}$	$\frac{N_{exit}}{N_{city \times ind}}$	$\frac{N_{entry} - N_{exit}}{N_{city \times ind}}$
	(1)	(2)	(3)	(4)
$ln(UMA)$	0.143*** (0.037)	0.048 (0.033)	-0.322*** (0.044)	0.370*** (0.059)
$ln(EUMA)$	0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	0.001 (0.001)
Dependent Variable's Mean	0.197	0.158	0.121	0.037
Time-varying city controls	Yes	Yes	Yes	Yes
City-industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	529.169	529.169	529.169	529.169
K-P rk Wald F statistic	408.694	408.694	408.694	408.694
Observations	109,558	109,558	109,558	109,558

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable in Column (1) is exporting rate measured by the proportion of exporting firms in the total number of firms at the city-industry level. The dependent variables in Columns (2) and (3) are entry rate and exit rate, measured by the proportion of new entrant firms in the current year and firms exiting the market in subsequent years, respectively. The dependent variable in Column (4) is the net entry rate of firms which is the difference of dependent variables between Columns (2) and (3).

Table 12: Lag Effects of Upstream Market Access

Dependent Variables:	<i>ln(export)</i>			
	(1)	(2)	(3)	(4)
<i>ln(UMA)</i>	0.083*** (0.029)			
<i>ln(UMA)_lag1</i>		0.019 (0.029)		
<i>ln(UMA)_lag2</i>			0.019 (0.030)	
<i>ln(UMA)_lag3</i>				-0.054 (0.034)
<i>ln(EUMA)</i>	-0.005 (0.008)	-0.005 (0.008)	-0.011 (0.009)	-0.006 (0.010)
<i>ln(dist_port)</i>	-0.084 (0.053)	-0.094** (0.042)	-0.057 (0.058)	-0.033 (0.061)
Dependent Variable's Mean	13.907	14.263	14.448	14.614
Time-varying firm controls	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	70.242	42.597	37.998	28.584
K-P rk Wald F statistic	1347.183	1361.008	909.363	550.841
Observations	268,168	170,787	116,328	71,113

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors are clustered at the city-industry level and shown in parenthesis. b) The dependent variable is the log of exports at the firm level. *ln(UMA)_lag1*, *ln(UMA)_lag2* and *ln(UMA)_lag3* represent the lagged *ln(UMA)* by one, two and three periods, respectively.

Table 13: Robustness Checks

Dependent Variables:	$\ln(\text{export})$				
	(1)	(2)	(3)	(4)	(5)
$\ln(UMA)$	0.084*** (0.029)	0.078*** (0.029)	0.086*** (0.028)	0.076** (0.035)	
$\ln(EUMA)$	-0.005 (0.008)	-0.000 (0.009)	-0.010 (0.008)	-0.006 (0.008)	-0.005 (0.008)
$\ln(\text{dist_port})$	-0.086 (0.053)	-0.016 (0.064)	-0.034 (0.050)	0.019 (0.042)	-0.084 (0.053)
cost	-0.002*** (0.000)				
$\ln(\text{cluster})$			0.239*** (0.059)		
$\ln(UMA^{\text{output}})$					0.080*** (0.028)
Dependent Variable's Mean	13.908	13.898	13.907	13.854	13.907
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes	Yes
Tariff-year controls	No	Yes	No	No	No
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	70.262	52.114	72.657	10.432	70.538
K-P rk Wald F statistic	1347.968	5175.899	1346.582	173.396	1351.696
Observations	268,001	243,984	268,137	148,816	268,168

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of exports at the firm level. Column (1) excludes the firms in the 36 major Chinese cities as described in Section 2.3. Column (2) presents the estimation based on the balanced panel sample, which only includes firms that export consistently throughout the sample period. Column (3) controls $\ln(\text{cluster})$, the logs of cluster index at city level. Column (4) substitutes the baseline upstream market access with UMA^{output} which is constructed based on upstream firms' total output rather than value added.

Appendix For Online Publication

A Construction of the Railway Network Database

As the basis for the empirical analysis, we construct a panel dataset of China’s railway network and calculate the shortest railway distances between all pairs of prefecture-level cities from 1998 to 2021. These distances are subsequently used to construct the upstream market access (UMA) measure, which captures firms’ access to domestic intermediate suppliers.

We begin by constructing vector maps of China’s railway network. The primary source is the printed *China Railway Yearbook* published annually by the Ministry of Railways of China.³⁰ The yearbooks report operating railway lines and railway projects under construction in each year. Owing to map-scale limitations, minor feeder lines serving county-level and lower-level administrative districts are omitted. Since our analysis focuses on connectivity between prefecture-level cities, these omissions have negligible effects on regional accessibility. We digitize these printed maps by manually drawing the outlines of each railway line with the aid of geographic information software (ArcGIS). Specifically, we use 2016 as the benchmark year, first digitizing the complete railway network for that year in ArcGIS and then updating the network backward and forward using the corresponding yearbooks.

The second step is to identify railway stations. We obtain the 2016 national railway station list from CNRDS, which contains the geographic coordinates of 7,042 stations.³¹ We convert these records into geographic location points and match them to the railway network. We then use Python to collect the opening years of more than 5,000 stations from Baidu Baike,³² while manually complementing the remaining stations and those opened

³⁰These documents are printed publications. They can be accessed at <https://cyfd.cnki.com.cn/N2020010195.htm>.

³¹The railway network constructed in this paper covers mainland China and excludes Hong Kong, Macau, and Taiwan.

³²Baidu Baike is a Chinese online encyclopedia covering a wide range of topics.

after 2016 using railway yearbooks and newspaper reports. This procedure enables us to identify the stations operating in each year.

Next, we match prefecture-level cities to the railway network. Different from Donaldson and Hornbeck (2016), who connect each city center to its nearest railway line, we combine information on administrative boundaries and railway stations. If a prefecture contains a single railway station, the city is matched to that station. If multiple stations exist, the city is connected to all of them. This approach reflects the fact that different stations within the same city often serve different railway corridors.³³

We then distinguish between conventional railways and high-speed railways (HSR). Only conventional railways carry freight and therefore determine transportation costs for goods and intermediate inputs, whereas HSR primarily transports passengers and facilitates the movement of labor and information. Accordingly, we construct separate network maps for conventional railways (1998–2021) and HSR (2008–2021). Figure A1 illustrates the conventional railway networks in 1998 and 2021. Railway density increased substantially during the sample period, particularly in eastern and southern China, while the western region also experienced considerable network expansion. The HSR network largely follows the conventional railway network and is considerably denser in eastern China by 2021.

Finally, we calculate the shortest railway distance between every pair of prefecture-level cities using the shortest-path algorithm of Dijkstra et al. (1959). Each city is treated as a node in the railway network, and the actual railway distance between adjacent nodes serves as the edge weight. Starting from an origin city, the algorithm recursively updates the shortest distance to all reachable destinations until the shortest-path tree is obtained. Applying this procedure to every prefecture yields a complete panel of pairwise railway distances over time.

³³For example, Beijing West Railway Station and Beijing South Railway Station serve different directions of the national railway network. Restricting Beijing to only one station would artificially lengthen the shortest railway distance to certain destinations.

Several caveats regarding the railway database deserve mention. First, manual digitization and station matching inevitably introduce some measurement error. We minimize this problem by using as many auxiliary location points and ArcGIS topology tools as possible. Second, the railway distances derived from the digitalized network may differ slightly from the actual operating distances. Our analysis, however, relies on relative differences in railway accessibility across cities and their evolution over time rather than exact railway distances. Using a common benchmark year ensures consistency across both space and time. Finally, because our focus is on interregional connectivity, we abstract from the within-city distance between the city center and railway stations, which is typically covered by local road transportation.

B Derivation of Equilibrium Conditions and Proofs

B1 Free Entry Equilibrium in a Closed Economy

Our steps to the solution of the free-entry equilibrium follow Melitz and Ottaviano (2008) and Mayer, Melitz, and Ottaviano (2014). This section shows the derivation of the following equilibrium quantities: the measure of varieties produced (μ), average price ($\bar{p}$), the measure of entrant firms in the economy (N_E), the measure of surviving firms in equilibrium (N).

In the closed economy, the free-entry equilibrium condition in Equation (15) can be rewritten as

$$\begin{aligned}
\int_0^{c_M} \sum_{m=0}^{M(c_0)-1} \pi(m, c_0) dG(c_0) &= f_E \quad \Rightarrow \\
\int_0^{c_M} \sum_{m \leq m(c_0, c_M)} \pi(m, c_0) dG(c_0) &= f_E \quad \Rightarrow \\
\sum_{m=0}^{\infty} \left[\int_0^{\omega^m c_M} \pi(m, c_0) dG(c_0) \right] &= f_E, \tag{57}
\end{aligned}$$

where

$$\pi(m, c_0) = \frac{L}{4\gamma} (c_M - \omega^{-m} c_0)^2.$$

Because the parameterization of $G(c_0)$ implies that

$$dG(c_0) = \frac{\xi c_0^{\xi-1}}{\bar{c}^\xi} dc_0,$$

we can solve equation (57) for the closed-form solution of c_M :

$$c_M = \left(\frac{\gamma\theta}{L\zeta} \right)^{\frac{1}{\xi+2}} \quad (58)$$

where $\theta = 2(\xi + 1)(\xi + 2)(\bar{c})^\xi f_E$ and $\zeta = \frac{1}{1-\omega^\xi}$.

Next, we rewrite the measure of varieties produced, μ , and their average price $\bar{p}$ as functions of c_M , and combine them with equation (58) to obtain their solutions.

(i) Expressing μ as a function of c_M . using the definition of the cutoff survival cost

$$c_M = \frac{\alpha\gamma + \eta\mu\bar{p}}{\eta\mu + \gamma},$$

we have

$$\mu = \frac{\gamma(\alpha - c_M)}{\eta(c_M - \bar{p})}.$$

(ii) Expressing $\bar{p}$ as a function of c_M . Because $p(v) = \frac{1}{2}(c_M + v)$ for any v in equilibrium, it holds that

$$\bar{p} = \frac{1}{2}(c_M + \bar{v}),$$

where $\bar{v}$ is the average marginal cost of all varieties produced.

Defining $M_v(c)$ as the measure of varieties that could be produced by the economy with production costs lower than c , the average cost can be written as

$$\bar{v} = \frac{1}{\mu} \int_0^{c_M} v dM_v(v).$$

(ii) Expressing N_E as a function of c_M . Given N_E , the measure of entrant firms in the economy, the measure of varieties produced per unit of entrants is given by

$$H_v(c_0) = \frac{M_v(c_0)}{N_E}.$$

In other words, for each entrant, $H_v(c)$ measures its probability of producing any potential variety with cost $v < c$, given its cost structure across varieties. Therefore,

$$\begin{aligned}
H_v(c_0) &= \sum_{m=0}^{\infty} G(\omega^m c_0) \\
&= \sum_{m=0}^{\infty} \omega^{mk} G(c_0) \\
&= \frac{1}{1 - \omega^\xi} G(c_0) \\
&= \zeta G(c_0).
\end{aligned} \tag{59}$$

The measure of varieties produced in equilibrium equals the measure produced by the N_E entrants with a real cutoff c_M , namely,

$$\mu = N_E H_v(c_M) = N_E \zeta G(c_M).$$

Now we obtain the solution for $\bar{v}$:

$$\begin{aligned}
\bar{v} &= \frac{1}{\mu} \int_0^{c_M} v dM_v(v) \\
&= \frac{1}{N_E H_v(c_M)} \int_0^{c_M} v N_E dH_v(v) \\
&= \frac{1}{\zeta G(c_M)} \int_0^{c_M} v \zeta dG(v) \\
&= \frac{1}{G(c_M)} \int_0^{c_M} v dG(v) \\
&= \frac{\xi}{\xi + 1} c_M.
\end{aligned} \tag{60}$$

Using the last line, we obtain the solutions for $\bar{p}$, μ , and N_E :

$$\bar{p} = \frac{1}{2}(c_M + \bar{v}) = \frac{2\xi + 1}{2(\xi + 1)} c_M, \tag{61}$$

$$\mu = \frac{\gamma(\alpha - c_M)}{\eta(c_M - \bar{p})} = \frac{2(\xi + 1)\gamma}{\eta} \frac{\alpha - c_M}{c_M}, \tag{62}$$

$$N_E = \frac{M}{H_v(c_M)} = \frac{2(\xi + 1)\gamma}{\eta\zeta} \frac{\alpha - c_M}{c_M} \left(\frac{\bar{c}}{c_M} \right)^\xi. \tag{63}$$

(iv) Finally, the measure of surviving firms in equilibrium equals the number of entrants whose core costs are below c_M :

$$N = N_E G(c_M) = \frac{2(\xi + 1)\gamma}{\eta\zeta} \frac{\alpha - c_M}{c_M}. \quad (64)$$

B2 Free Entry Equilibrium in an Open Economy

Consider a potential entrant located in city kg of country k . Its expected profit is the sum of the profits obtained from all destination markets. Reordering the integration first across firms producing variety m and then across varieties, the free-entry condition can be written as

$$\sum_{j=1}^J \sum_{f=1}^{F_j} \sum_{m=0}^{\infty} \left[\int_0^{\omega^m c_{M,jfkg}} \pi_{jfk g}(m, c_0) dG_{c,k}(c_0) \right] = f_E, \quad (65)$$

where

$$\pi_{jfk g}(m, c_0) = \frac{L_{jf}}{4\gamma} \left(c_{M,jfjf} - e^{\tau_{jfk g}} \omega^{-m} c_0 \right)^2. \quad (66)$$

Given

$$G_{c,k}(c_0) = \left(\frac{c_0}{\bar{c}_k} \right)^{\xi}, \quad dG_{c,k}(c_0) = \frac{k c_0^{k-1}}{\bar{c}_k^{\xi}} dc_0,$$

the expected profit obtained from destination jf is

$$\begin{aligned} & \sum_{m=0}^{\infty} \int_0^{\omega^m c_{M,jfkg}} \pi_{jfk g}(m, c_0) dG_{c,k}(c_0) \\ &= \frac{\zeta L_{jf} \rho_{jfk g} \bar{c}_{M,jfjf}^{\xi+2}}{2\gamma(\xi+1)(\xi+2)\bar{c}_k^{\xi}}. \end{aligned} \quad (67)$$

The free-entry condition for city kg becomes

$$\sum_{j=1}^J \sum_{f=1}^{F_j} \rho_{jfk g} L_{jf} \bar{c}_{M,jfjf}^{\xi+2} = \frac{\gamma \theta_i}{\zeta}, \quad (68)$$

where $\theta_i = 2(\xi+1)(\xi+2)f_E \bar{c}_k^{\xi}$.

Collecting the free-entry conditions for all cities gives

$$\mathcal{P}\mathbf{z} = \frac{\gamma}{\zeta}\boldsymbol{\theta}, \quad (69)$$

where $\mathcal{P}$ and $\boldsymbol{\theta}$ are defined in Section 2.2, and $\mathbf{z}$ is the vector of $z_{jf} \equiv L_{jf}c_{M,jfff}^{\xi+2}$.

For each city jf , let $\mathcal{P}_{jf}^{\boldsymbol{\theta}}$ denote the matrix obtained by replacing the column of $\mathcal{P}$ corresponding to city jf with $\boldsymbol{\theta}$. Cramer's rule gives

$$c_{M,jfff} = \left(\frac{\gamma}{L_{jf}\zeta} \frac{|\mathcal{P}_{jf}^{\boldsymbol{\theta}}|}{|\mathcal{P}|} \right)^{\frac{1}{\xi+2}}. \quad (70)$$

At any destination market jf , the distribution of delivered marginal costs has the same Pareto shape parameter ξ . The average delivered marginal cost, average price, and aggregate measure of varieties sold in that market are therefore

$$\bar{v}_{jf} = \frac{\xi}{\xi+1} c_{M,jfff}, \quad (71)$$

$$\bar{p}_{jf} = \frac{2\xi+1}{2(\xi+1)} c_{M,jfff}, \quad (72)$$

$$\mu_{jf} = \frac{2(\xi+1)\gamma}{\eta} \frac{\alpha - c_{M,jfff}}{c_{M,jfff}}. \quad (73)$$

For firms originating in country k , define the normalized measure of varieties associated with a unit mass of entrants as

$$H_{v,k}(c) = \sum_{m=0}^{\infty} G_{c,k}(\omega^m c) = \zeta G_{c,k}(c) = \zeta \left(\frac{c}{\bar{c}_k} \right)^{\xi}. \quad (74)$$

The total measure of varieties available in destination market jf is consequently

$$\begin{aligned} \mu_{jf} &= \sum_{k=1}^J \sum_{g=1}^{F_i} N_{E,k} H_{v,k}(c_{M,jfkg}) \\ &= \zeta c_{M,jfff}^{\xi} \sum_{k=1}^J \sum_{g=1}^{F_i} \rho_{jfkg} \frac{N_{E,k}}{\bar{c}_k^{\xi}}. \end{aligned} \quad (75)$$

Define

$$u^{kg} \equiv \frac{N_{E,k}}{\bar{c}_k^{\xi}}, \quad b_{jf} \equiv \frac{\mu_{jf}}{\zeta c_{M,jfff}^{\xi}} = \frac{2(\xi+1)\gamma}{\eta\zeta} \frac{\alpha - c_{M,jfff}}{c_{M,jfff}^{\xi+1}}. \quad (76)$$

Collecting these terms in the city-ordered vectors $\mathbf{u}$ and $\mathbf{b}$, Equation (75) becomes

$$\mathcal{P}^\top \mathbf{u} = \mathbf{b}. \quad (77)$$

Let $\mathcal{C}_{kg,jf}$ denote the cofactor of the element ρ_{jfk_g} in $\mathcal{P}$. The equilibrium mass of entrants in city kg is

$$\begin{aligned} N_{E,kg} &= \bar{c}_k^\xi \sum_{j=1}^J \sum_{f=1}^{F_j} \frac{\mathcal{C}_{kg,jf}}{|\mathcal{P}|} b_{jf} \\ &= \frac{2(\xi+1)\gamma\bar{c}_k^\xi}{\eta\zeta} \sum_{j=1}^J \sum_{f=1}^{F_j} \frac{\mathcal{C}_{kg,jf}}{|\mathcal{P}|} \frac{\alpha - c_{M,jfjf}}{\bar{c}_{M,jfjf}^{\xi+1}}. \end{aligned} \quad (78)$$

Finally, the mass of firms headquartered in city kg that survive in their local market is

$$N^{kg} = N_{E,kg} G_{c,k}(c_{M,kgkg}) = N_{E,kg} \left(\frac{c_{M,kgkg}}{\bar{c}_k} \right)^\xi. \quad (79)$$

B3 Proof of Proposition 1

Fix the core productivity φ_0 of a firm located in country 1. Recall that

$$\frac{\partial \ln c_0}{\partial \tau_1} = a_1, \quad \frac{\partial \ln \theta_1}{\partial \tau_1} = \xi a_1, \quad \frac{\partial \ln \theta_2}{\partial \tau_1} = \xi \chi a_1. \quad (80)$$

Since neither ρ_0 nor ρ_2 varies with τ_1 , differentiation of κ_2 gives

$$\frac{\partial \kappa_2}{\partial \tau_1} = \xi a_1 \kappa_5, \quad \frac{\partial \ln c_{M,2s2s}}{\partial \tau_1} = a_1 \Delta. \quad (81)$$

Moreover,

$$\kappa_2 - \kappa_5 = (1 - \chi)\theta_2(1 + \rho_0^2 \rho_2) > 0. \quad (82)$$

Since $\kappa_2 > 0$, Equation (82) implies $\Delta < \xi/(\xi+2) < 1$. If $\kappa_5 > 0$, it additionally implies $0 < \Delta < 1$.

Recall the continuous export-scope index h_{1f2s} , $f = s, l$. Because $c_{M,1s2s} = e^{-\tau_0} c_{M,2s2s}$ and $c_{M,1l2s} = e^{-(\tau_0+\tau_1)} c_{M,2s2s}$, Equations (80)–(81) imply

$$\frac{\partial h_{1s2s}}{\partial \tau_1} = \frac{a_1(\Delta - 1)}{\ln \omega^{-1}} < 0, \quad \frac{\partial h_{1l2s}}{\partial \tau_1} = \frac{a_1(\Delta - 1) - 1}{\ln \omega^{-1}} < 0. \quad (83)$$

For seaboard firms, $a_1\Delta$ is the response of the export cutoff generated through destination-market competition, while a_1 is the response of the firm's unit cost. Since $\Delta < 1$, the cost-saving effect dominates. For inland firms, the additional term $-1/\ln \omega^{-1}$ is the direct final-goods transportation effect.

For completeness, define $h_{1f2l} = [\ln c_{M,1f2l} - \ln c_0]/\ln \omega^{-1}$ for exports to city $2l$. Since $\partial \ln c_{M,2l2l}/\partial \tau_1 = \xi \chi a_1/(\xi + 2)$, the corresponding responses are

$$\frac{\partial h_{1s2l}}{\partial \tau_1} = \frac{a_1 \left(\frac{\xi}{\xi+2} \chi - 1 \right)}{\ln \omega^{-1}} < 0, \quad \frac{\partial h_{1l2l}}{\partial \tau_1} = \frac{a_1 \left(\frac{\xi}{\xi+2} \chi - 1 \right) - 1}{\ln \omega^{-1}} < 0. \quad (84)$$

For either destination $2g$, $g = s, l$, the number of exported varieties is

$$M_{1f2g}(\varphi_0, \tau_1) = \max \{0, 1 + \lfloor h_{1f2g}(\varphi_0, \tau_1) \rfloor\}. \quad (85)$$

A reduction in τ_1 raises every continuous scope index in Equations (83)–(84). Since Equation (85) is weakly increasing in its argument, export scope weakly increases for both seaboard and inland firms and increases strictly whenever the relevant continuous index crosses an integer threshold. ■

B4 Proof of Proposition 2

Fix a firm's core-productivity draw φ_0 and an existing exported variety m . For compactness, write $v_m \equiv v(m, c_0)$, $C \equiv c_{M,2s2s}$, $d_s \equiv e^{\tau_0} v_m$, and $d_l \equiv e^{\tau_0+\tau_1} v_m$. From Equations (40) to (43),

$$\frac{\partial v_m}{\partial \tau_1} = a_1 v_m, \quad \frac{\partial C}{\partial \tau_1} = a_1 \Delta C. \quad (86)$$

Product-level export revenues are

$$r_{1s2s}(m, c_0) = \frac{L_{2s}}{4\gamma}(C^2 - d_s^2), \quad r_{1l2s}(m, c_0) = \frac{L_{2s}}{4\gamma}(C^2 - d_l^2). \quad (87)$$

Because $\partial d_s / \partial \tau_1 = a_1 d_s$ and $\partial d_l / \partial \tau_1 = (1 + a_1) d_l$, differentiation gives

$$\begin{aligned} \frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} &= \frac{L_{2s}}{2\gamma} \left[\underbrace{a_1 \Delta C^2}_{\text{competition effect}} - \underbrace{a_1 d_s^2}_{\text{intermediate-input cost effect}} \right], \\ \frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} &= \frac{L_{2s}}{2\gamma} \left[\underbrace{a_1 \Delta C^2}_{\text{competition effect}} - \underbrace{a_1 d_l^2}_{\text{intermediate-input cost effect}} - \underbrace{d_l^2}_{\text{final-goods transportation effect}} \right]. \end{aligned} \quad (88)$$

(89)

These expressions are derivatives with respect to an increase in τ_1 . Hence, when τ_1 falls, the positive competition term represents a revenue loss caused by the lower destination-market cutoff, whereas each negative cost term represents a revenue gain.

Using $c_{M,1s2s} = e^{-\tau_0} C$, Equation (88) becomes

$$\frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} = \frac{a_1 L_{2s} e^{2\tau_0}}{2\gamma} (\Delta c_{M,1s2s}^2 - v_m^2). \quad (90)$$

Therefore,

$$\frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} \leq 0 \iff \frac{v_m}{c_{M,1s2s}} \geq \Delta^{1/2} = \left(\frac{\xi}{\xi + 2} \frac{\kappa_5}{\kappa_2} \right)^{1/2}. \quad (91)$$

For inland firms, define $\nu \equiv a_1^{-1} > 0$. Since $c_{M,1l2s} = e^{-(\tau_0 + \tau_1)} C$, Equation (89) becomes

$$\frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} = \frac{a_1 L_{2s} e^{2(\tau_0 + \tau_1)}}{2\gamma} [\Delta c_{M,1l2s}^2 - (1 + \nu) v_m^2]. \quad (92)$$

It follows that

$$\frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} \leq 0 \iff \frac{v_m}{c_{M,1l2s}} \geq \left(\frac{\Delta}{1 + \nu} \right)^{1/2} = \left(\frac{\xi}{\xi + 2} \frac{\kappa_5}{(1 + \nu) \kappa_2} \right)^{1/2}. \quad (93)$$

Finally, delivered markups satisfy

$$\lambda_{1s2s}(m, c_0) = \frac{C}{2} \left(1 - \frac{v_m}{c_{M,1s2s}} \right), \quad \lambda_{1l2s}(m, c_0) = \frac{C}{2} \left(1 - \frac{v_m}{c_{M,1l2s}} \right). \quad (94)$$

Substitution into Equations (91) and (93) yields

$$\frac{\partial r_{1s2s}(m, c_0)}{\partial \tau_1} \leq 0 \iff \lambda_{1s2s}(m, c_0) \leq \frac{C}{2} \left(1 - \Delta^{1/2}\right), \quad (95)$$

$$\frac{\partial r_{1l2s}(m, c_0)}{\partial \tau_1} \leq 0 \iff \lambda_{1l2s}(m, c_0) \leq \frac{C}{2} \left[1 - \left(\frac{\Delta}{1 + \nu}\right)^{1/2}\right]. \quad (96)$$

Thus, under $0 < \Delta < 1$, a reduction in τ_1 raises export revenues for lower-markup varieties and lowers them for higher-markup varieties. Since $\nu > 0$, the final-goods transportation effect raises the markup threshold for inland firms and therefore expands the set of their existing export varieties whose revenues increase.

■

B Additional Figure and Tables

Table C1: List of Major Port Cities

Port	Province	Port	Province
DaLian	LiaoNing	NingBo	ZheJiang
YingKou	LiaoNing	ZhouShan	ZheJiang
QinHuangDao	HeBei	WenZhou	ZheJiang
TianJin	TianJin	FuZhou	FuJian
YanTai	ShanDong	XiaMen	FuJian
QingDao	ShanDong	ShanTou	GuangDong
RiZhao	ShanDong	ShenZhen	GuangDong
ShangHai	ShangHai	GuangZhou	GuangDong
LianYunGang	JiangSu	ZhuHai	GuangDong
NanTong	JiangSu	ZhanJiang	GuangDong
SuZhou	JiangSu	FangChengGang	GuangXi
ZhenJiang	JiangSu	HaiKou	Hainan
NanJing	JiangSu		

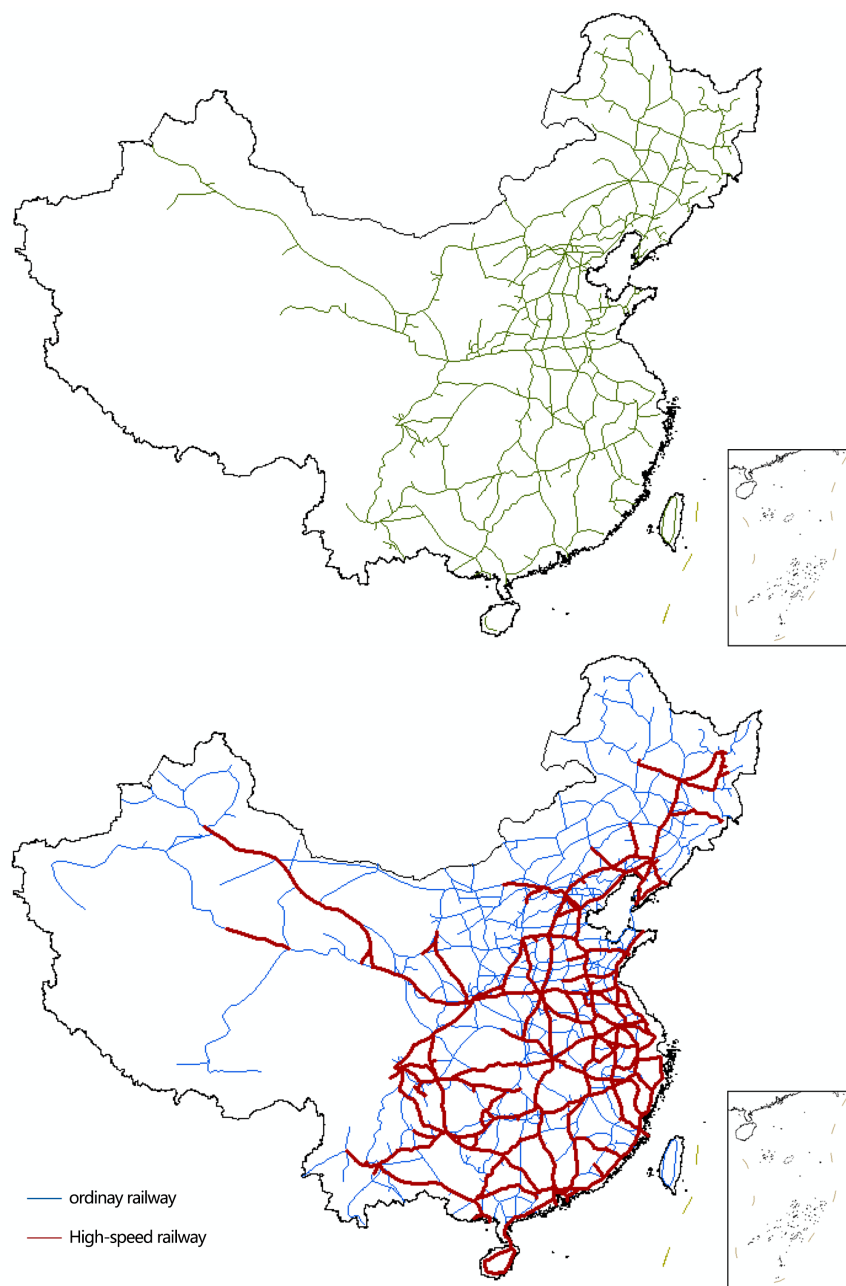


Figure A1: China's Main Railway Networks in 1998 and 2021

Source: authors' calculations.

Table C2: Upstream Market Access and Firm Exports (2SLS, First Stage)

Panel A. First stage for UMA		
Dependent Variables:	$\ln(UMA)$	
	(1)	(2)
$\ln(UMA)_{-geocost}$	0.982*** (0.002)	
$\ln(UMA)_{-straight}$		0.986*** (0.002)
Dependent Variable's Mean	13.907	13.907
Observations	268,168	268,168
Panel B. First stage for distance to ports		
Dependent Variables:	$\ln(dist_port)$	
	(1)	(2)
$\ln(dist_port)_{-geocost}$	0.968*** (0.019)	
$\ln(dist_port)_{-straight}$		0.679*** (0.052)
Dependent Variable's Mean	13.907	13.907
Observations	268,168	268,168
Time-varying firm controls	Yes	Yes
Time-varying city controls	Yes	Yes
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable in Panel A is the log of upstream market access and that in Panel B is the log of distance to the nearest port. $\ln(UMA)_{-geocost}$ and $\ln(dist_port)_{-geocost}$ are the logs of counterfactual UMA and $dist_port$ that are constructed by the artificial transportation network connected by the least costly paths. $\ln(UMA)_{-straight}$ and $\ln(dist_port)_{-straight}$ are the logs of counterfactual UMA and $dist_port$ that are constructed by the artificial transportation network connected by straight lines.

Table C3: Upstream Market Access and HS4-product-level Exports

Dependent Variables:	$\ln(\text{export}^{hs4})$			
	(1)	(2)	(3)	(4)
$\ln(UMA)$	-0.038 (0.032)	-0.036 (0.032)	-0.092** (0.041)	-0.050 (0.036)
$\ln(EUMA)$	0.022*** (0.007)	0.023*** (0.007)	0.023*** (0.007)	0.022*** (0.007)
$\ln(\text{dist_port})$	-0.069 (0.053)	-0.068 (0.053)	-0.067 (0.053)	-0.069 (0.053)
$\ln(UMA) \times \text{costr}$		0.005** (0.002)		
$\ln(UMA) \times \mathbb{1}(\text{costr}_{50th})$			0.114** (0.057)	
$\ln(UMA) \times \mathbb{1}(\text{costr}_{75th})$				0.049 (0.066)
Dependent Variable's Mean	11.658	11.658	11.658	11.658
Time-varying firm controls	Yes	Yes	Yes	Yes
Time-varying city controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	43.039	43.117	42.993	43.468
K-P rk Wald F statistic	1390.795	928.762	926.934	935.228
Observations	559,866	559,866	559,866	559,866

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the city-industry level and shown in parenthesis. The dependent variable is the log of exports at the HS4-product level. *costr* is firm's cost rate measured by the total costs per unit of revenue. $\mathbb{1}(\text{costr}_{50th})$ is an indicator variable for firms with cost rates higher than the 50th percentile, and $\mathbb{1}(\text{costr}_{75th})$ is an indicator those with cost rates higher than the 75th percentile.